



Fundamentals of MPRA Experiments

ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

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UNIVERSITÄT ZU LÜBECK



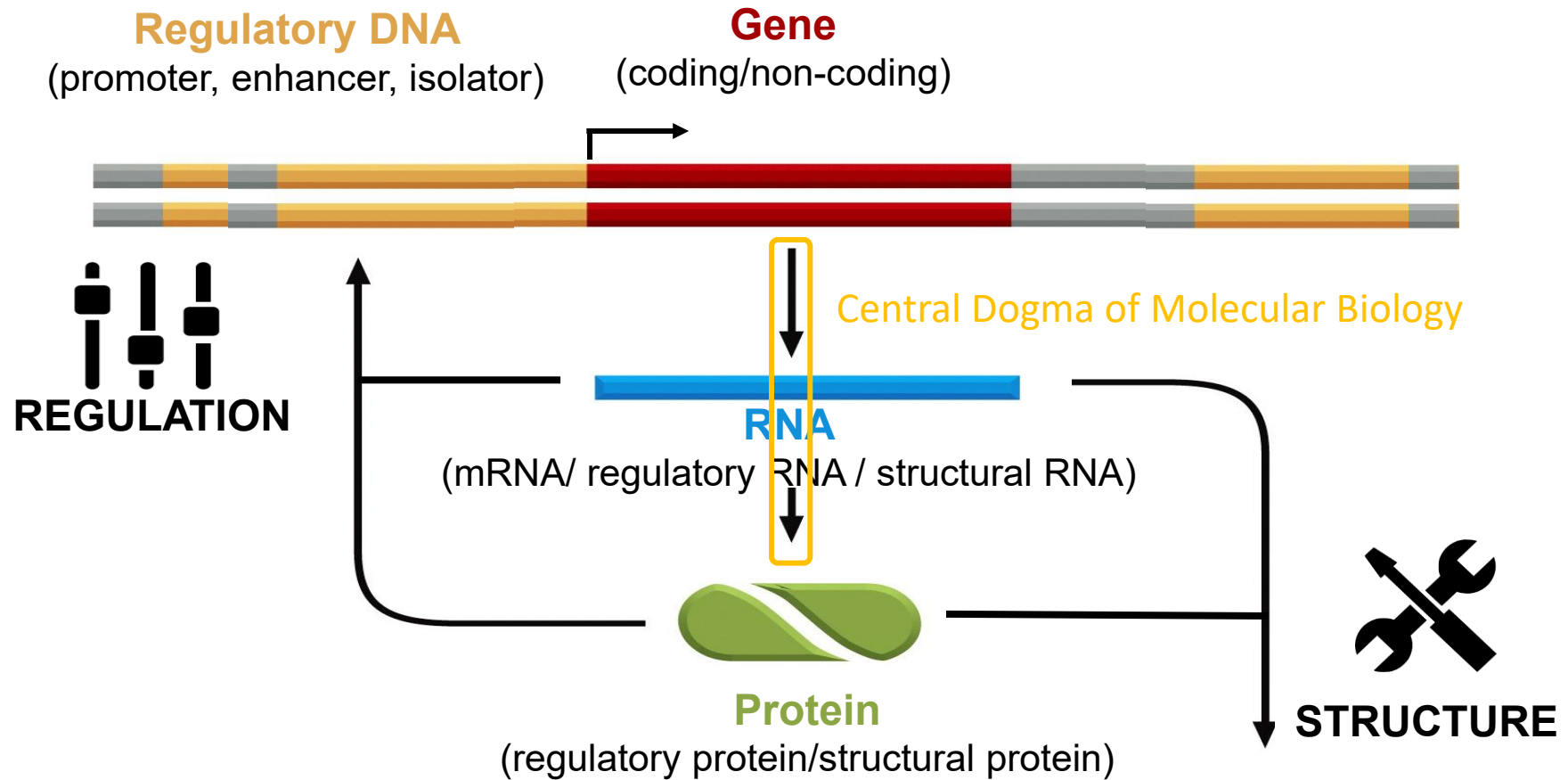
UNIVERSITÄTSKLINIKUM
Schleswig-Holstein



Berlin Institute
of Health
@Charité

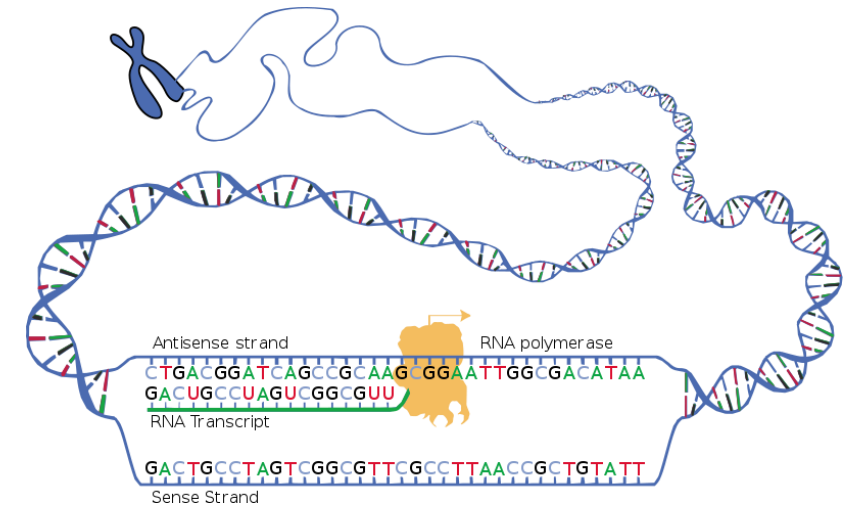
Part I – The underlying biology

Variant-to-function relation

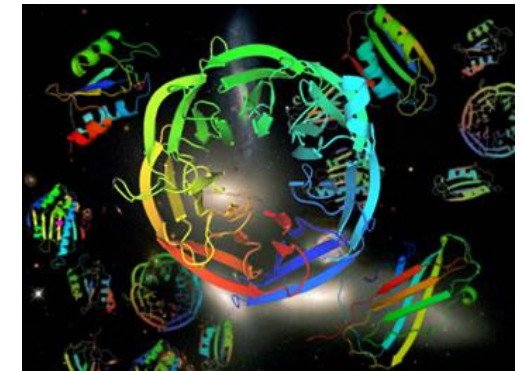
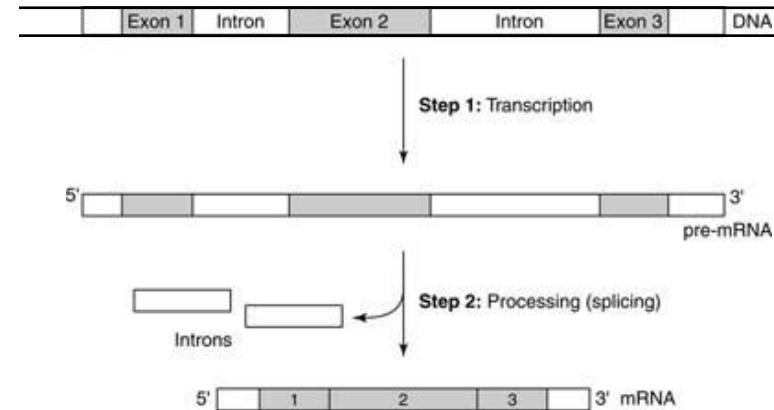


Complexity reminder...

- One gene many transcripts
- Exons/intron gene structures in eukaryotes
- Various steps of RNA processing
- RNA interference
- Cell-type specificity
- Protein modifications and complexes
- Protein protein interactions, catalytic activities, metabolite concentrations, ...
- ...



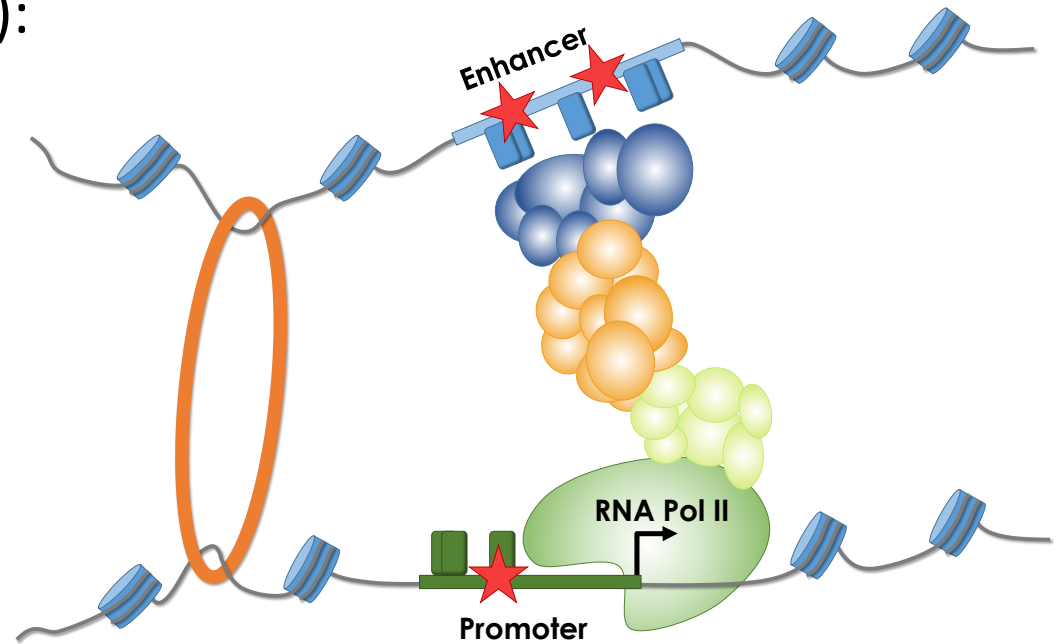
<https://www.genome.gov/genetics-glossary/Transcription>



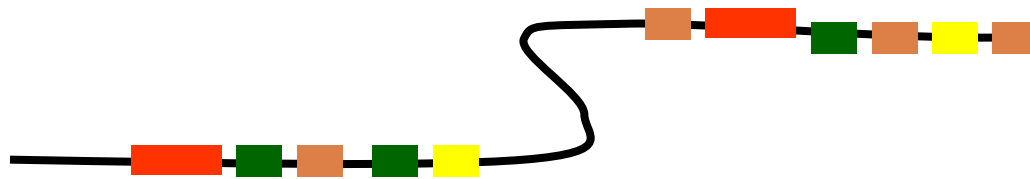
https://www2.lbl.gov/Science-Articles/Archive/sabl/2005/March/assets/proteins_as_galaxies.jpg

Two classes? Coding vs. Non-coding variants?

- **Non-coding sequences:** ~98% of the genome remaining after subtracting coding sequence
- **Functional non-coding elements:** like 3' and 5' UTRs, short and long RNA species, ...
- **Regulatory sequences** (~5-15% of genome):
enhancer, silencer, promoter, insulator
 - Open chromatin regions?
 - Chromatin associated?



TFBSs are clustered in promoters or in "sequence modules"



The distribution of binding sites in the genome is non-uniform

In small genomes, most sites are in promoters, and there is a bias toward a nucleosome free region near transcriptional start sites (TSSs)

In larger genomes (fly) CRMs (cis-regulatory-modules) exist which are frequently away from the TSS (enhancers)

A single binding site, without the context of other TF, is unlikely to represent a functional loci

Operational definitions of cis-regulatory modules

A promoter is the DNA sequence required for correct initiation of transcription

5' end of the gene (may overlap exons/coding sequence)

An enhancer is a DNA sequence that causes an increase in gene expression

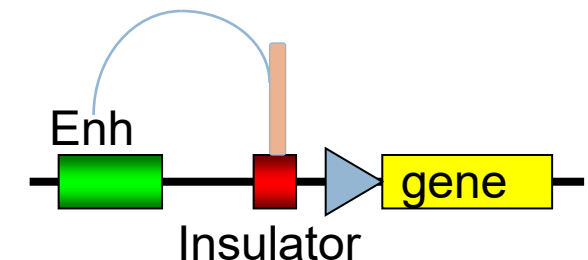
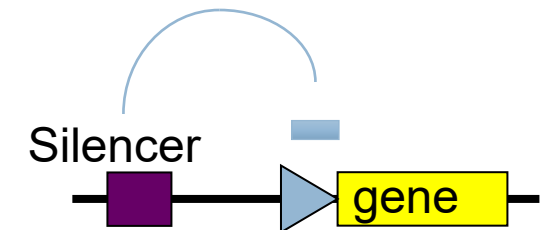
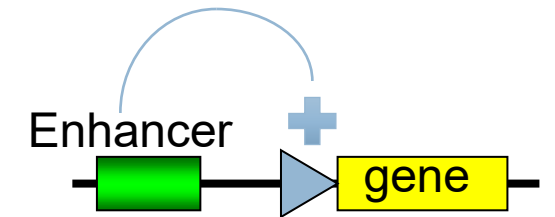
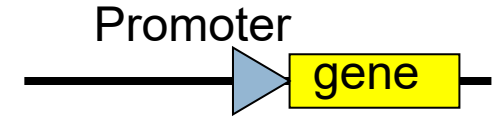
Act independently of position and orientation with respect to the gene

A silencer is a DNA sequence that causes a decrease in gene expression

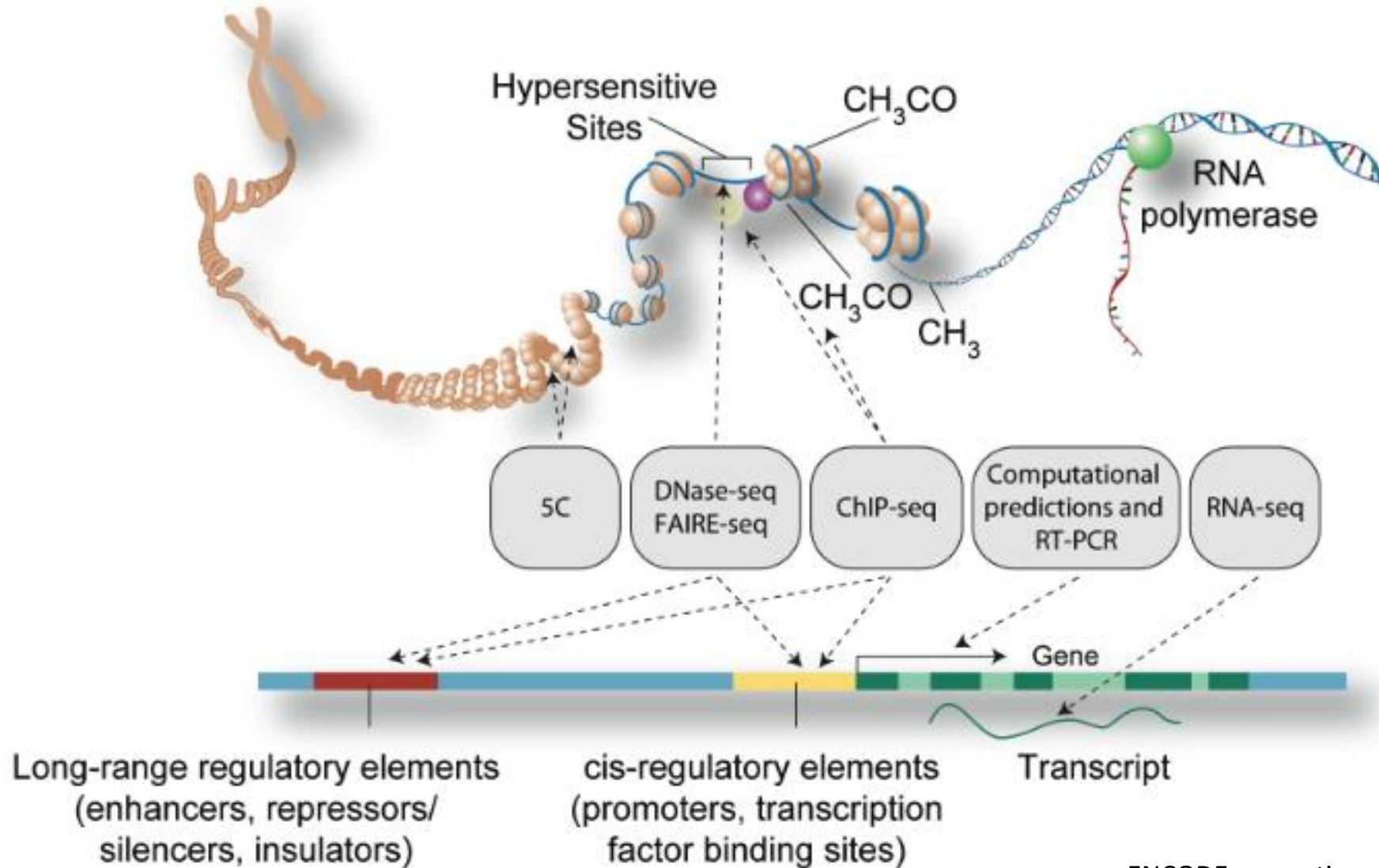
Similar to enhancer but has an opposite effect on gene expression (can be physically the same sequence and depend on cell-type)

An insulator is a DNA sequence that blocks activation of a promoter by an enhancer

Formed by topologically associated domains (TADs) by CTCF and Cohesin binding



Mapping Functional Elements



Diversity of evidence supporting regulatory regions



What is biological function?

“A genome segment may be biochemically active yet have no biologically meaningful function. An analogous situation arises when my shoe binds a piece of chewing gum during hot days in Houston. The binding of the chewing gum is an observable activity, but no reasonable person would claim that this is the function of shoes.”



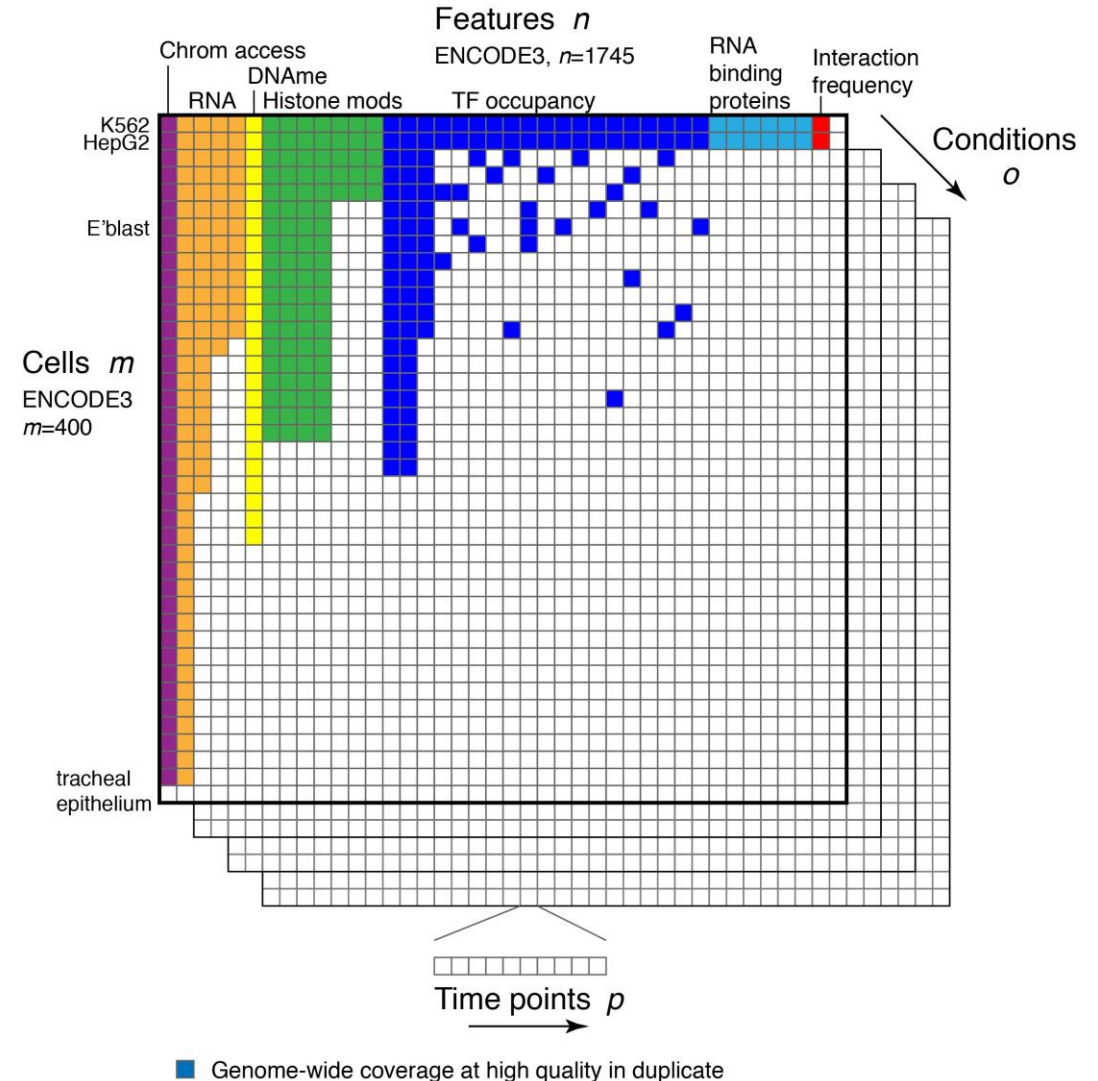
Dan Graur, 2016

The current state of mapping function-associated features

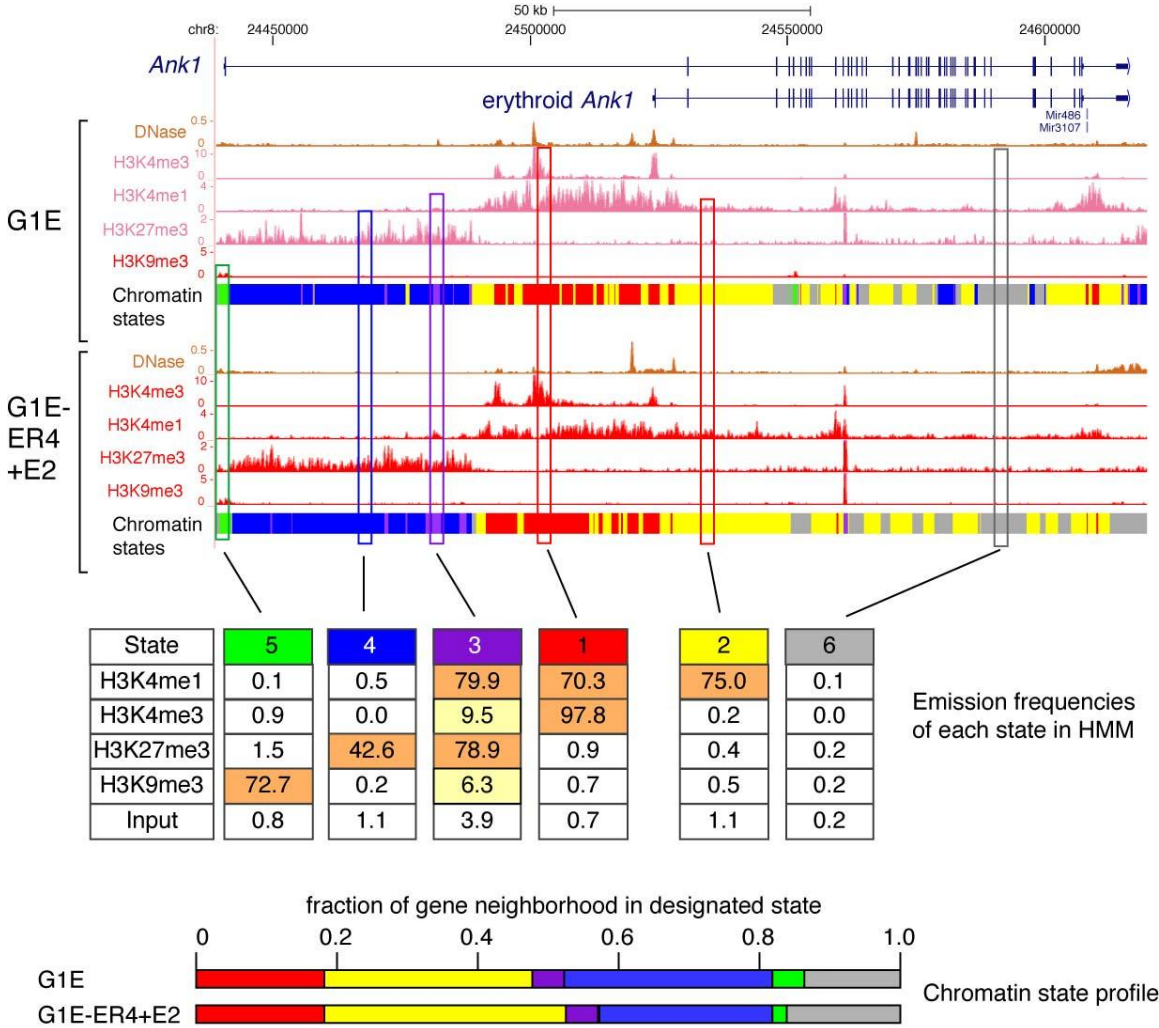
Thousands of experiments, but sparse across diversity of cell-types, conditions or developmental trajectories

Computational challenges:

- Data integration (batch/technical confounders)
- Data imputation
- Catalogs of elements



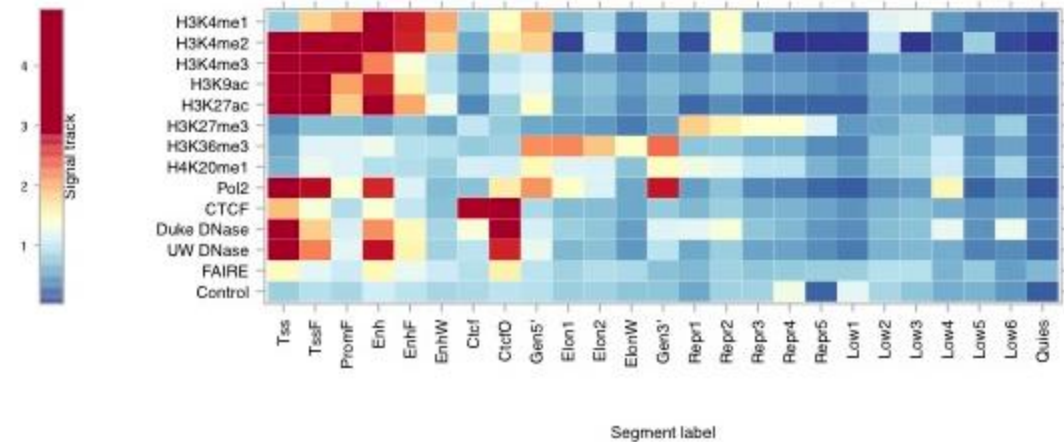
“Paint” genomic regions by dominant histone modifications: multivariate HMM (chromHMM)



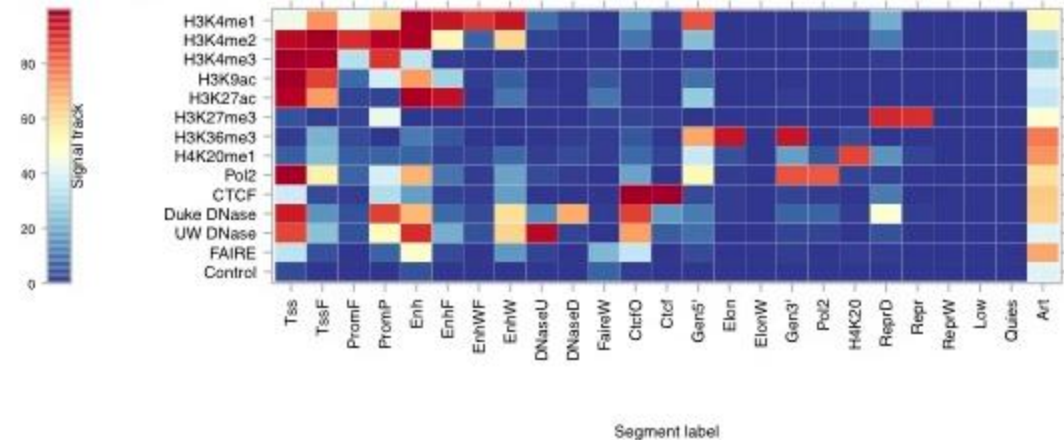
Integrate features using multivariate segmentations (Segway)

Regul	Regulatory
Tss	Transcription start site
Prom	Promoter
Gen5'	5' ends of genes
Elon	Elongation
Gen3'	3' ends of genes
Enh	Enhancer
EnhPr	Enhancer/Promoter
Ctcf	CTCF binding
Dnase	DNaseI sensitivity
Faire	FAIRE activity
Pol2	Pol2 binding
H4K20	H4K20 monomethylation
Repr	Repressed
Low	Low signal
Art	Artifact
Quies	Quiescent
W	Weak
F	Flanking
P	Poised
O	Open
D	Duke
U	U Washington

(A) Label mnemonics



(B) Segway



(C) ChromHMM

ENCODE Project Consortium (2012) Nature

M. Hoffman, J. Ernst et al. 2013. Integrative segmentations of function-associated marks. NAR

Catalogs of open chromatin regions

<https://screen.wenglab.org/>

EH38E2941902 chr11:5,267,726-5,267,987 D

[In Specific Biosamples](#) [Nearby Genomic Features](#) [TF and His-mod Intersection](#) [TF Motifs and Sequence Features](#) [Associated Gene Expression](#) [Associated RAMPAGE Signal](#) [Linked cCREs in other Assemblies](#)

[Functional Data](#) [EN-TE States](#) [Signal Profile](#) [Linked Genes](#)

Cell type agnostic classification

Cell Type	DNase max-Z	H3K4me3 max-Z	H3K27ac max-Z	CTCF max-Z	Group
cell type agnostic	1.68	2.19	2.02	1.45	distal enhancer-like signature
Total: 1					

Classification in Type A biosamples (all four marks available)

cell type	DNase Z-score	H3K4me3 Z-score	H3K27ac Z-score	CTCF Z-score	Group
K562	1.68	2.19	2.02	-0.01	distal enhancer-like signature
keratinocyte female donor ENCD0268AAA	0.54	0.16	-10.00	-10.00	low DNase

SCREEN

About Downloads Applets

Genomic Region in GRCh38

Enter a genomic region chr12:53,380,176-53,416,446

SCREEN

Search Candidate cis-Regulatory Elements by ENCODE

Search by Genomic Region in GRCh38

☒ chr.start-end ☐ Individual Inputs

Enter a genomic region chr12:53,380,176-53,416,446

What is SCREEN?
SCREEN is a web interface for searching and visualizing the Registry of candidate cis-Regulatory Elements (cCREs) derived from ENCODE data. The Registry contains 2,348,854 human cCREs in GRCh38 and 926,843 mouse cCREs in mm10, with homologous cCREs cross-referenced across species. SCREEN presents the data that support biochemical activities of the cCREs and the expression of nearby genes in specific cell and tissue types.

Version 4 Annotations:

Genome Assembly	cCRE Count	Cell/Tissue Types
Human (hg38)	2,348,854	1,888
Mouse (mm10)	926,843	382

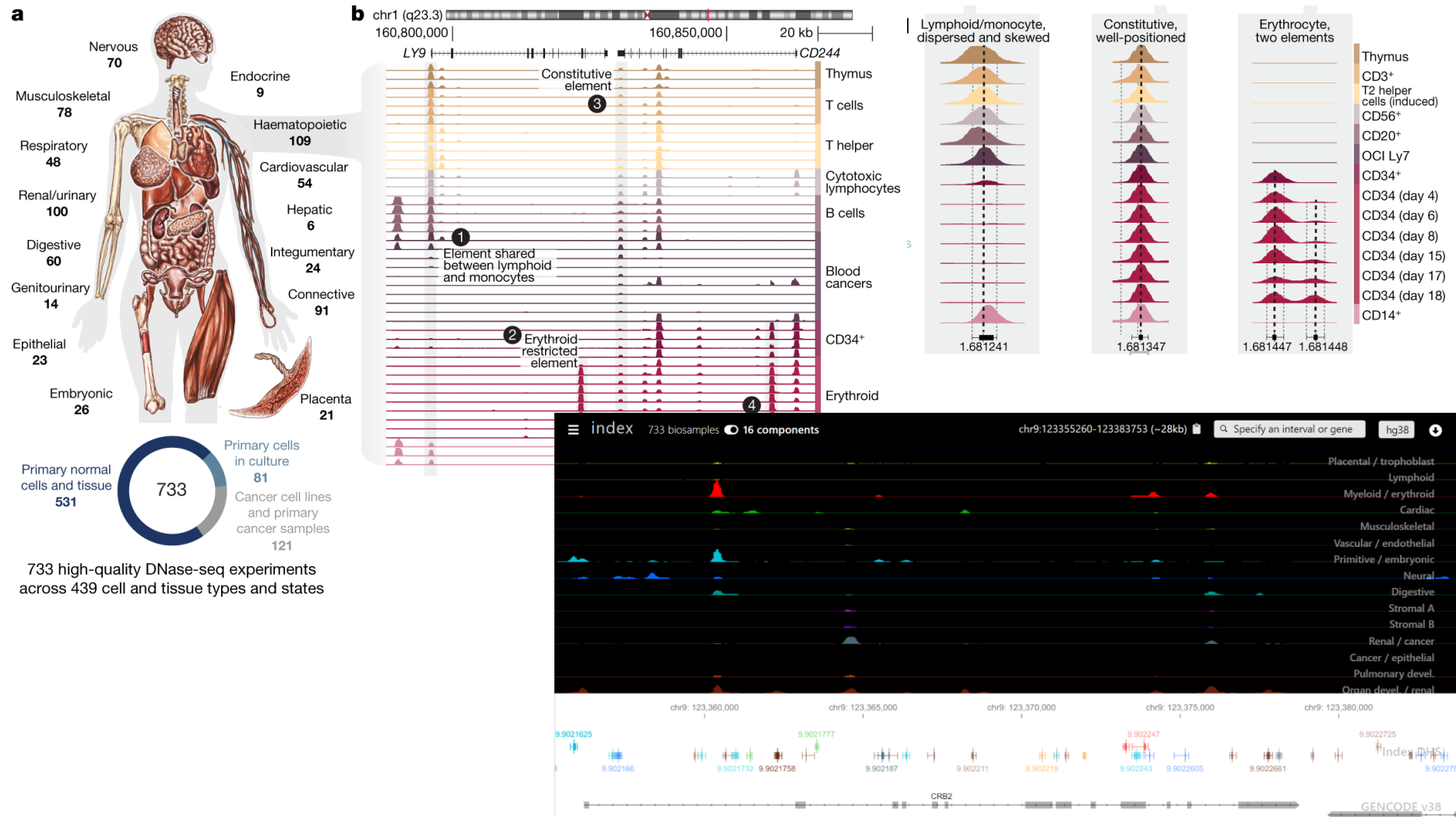
Copyright © Weng Lab, Moore Lab 2025.
How to Cite the ENCODE Encyclopedia, the Registry of cCREs, and SCREEN:
"ENCODE Project Consortium, et al. Nature 2020."

GRCh38:chr11

move <<< << < > >> >>> zoom in 1.5x 3x 10x zoom out 1.5x 3x 10x 100x

Catalogs of open chromatin regions (2)

<https://index.altius.org/>



Multiplex Assays of Variant Effects (MAVE)

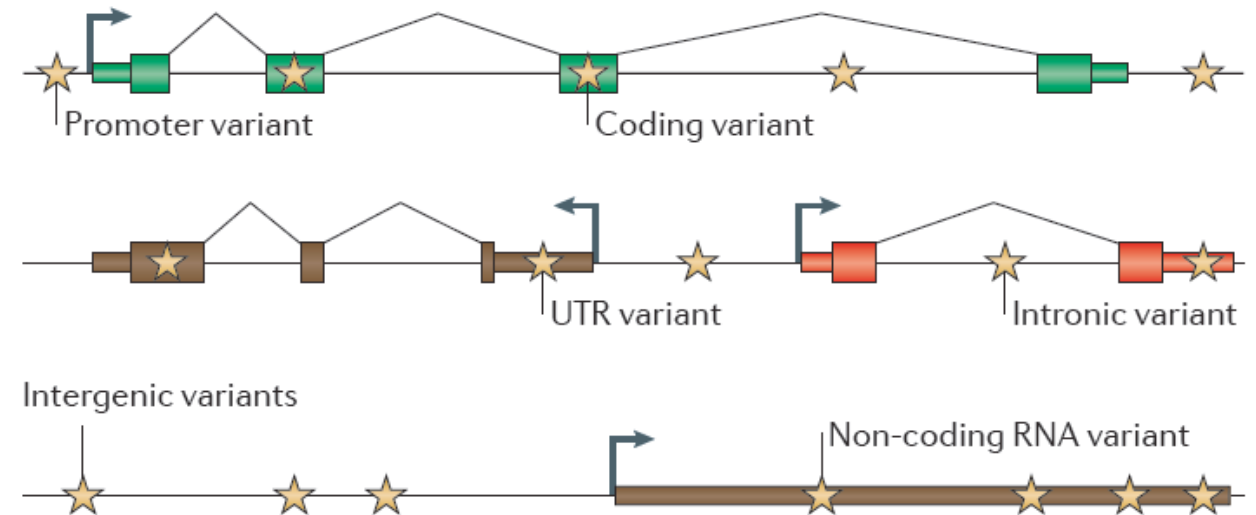
- A toolbox for interrogating the effects of sequence alterations

- Diverse molecular assays

- Lenti-virus, piggyBac transposon, ...
- Directed or random mutagenesis
- CRISPR/Cas and dCas fusions

- Diverse readouts

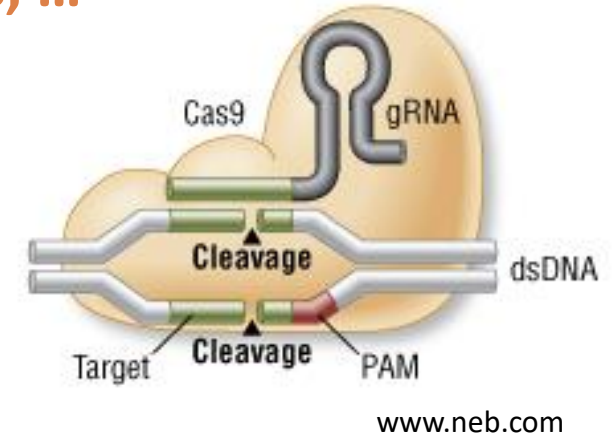
- Survival / proliferation
- Protein abundance
- Reporter signal (e.g. light), RNA expression, DNA and chromatin state



Modified from Cooper and Shendure, *Nature Review Genetics* 2011

Quantitative read-outs from regulatory sequences

- **CRISPR/Cas9: mutation, deletion, activator/repressor screens, ...**
- **Massively Parallel Reporter Assays (MPRAs) or similar:**
 - Test activity of regions (element catalogs / rules)
 - Measuring effects of genomically scattered mutations
 - Dense read outs for mutations in select regions

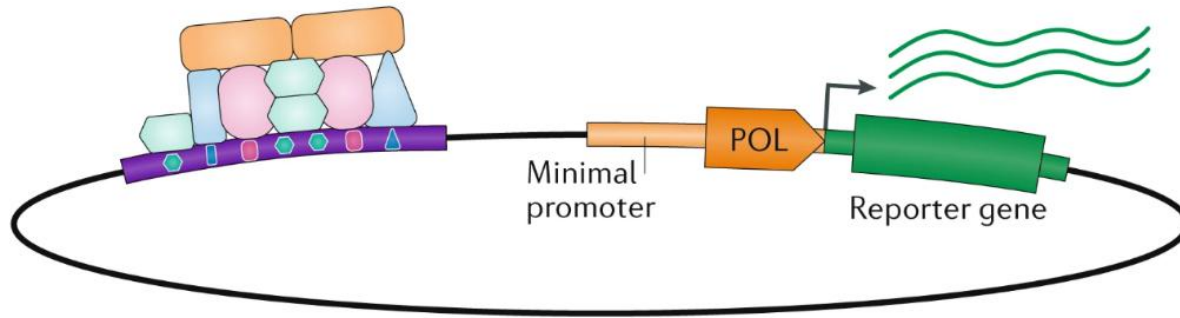


	Construct	Tested in	Detection	Advantages	Disadvantages
MPRA/ MPFD/ CRE-seq		Cell lines, Mouse liver, Mouse retina	Barcode RNA-seq	High BC multiplicity Quantitative	Episomal
STARR-seq		Cell lines	Enhancer RNA-seq	Quantitative	Low multiplicity Episomal
TRIP		Mouse ESCs	Barcode RNA-seq	Quantitative Genomic context	Low resolution

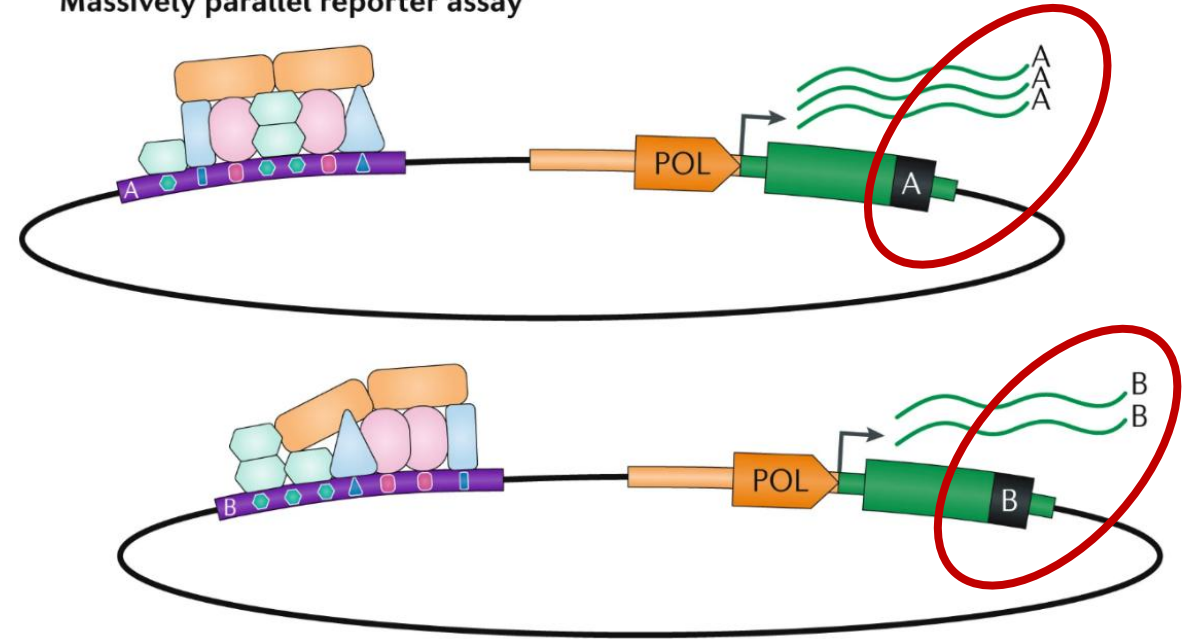
*F. Inoue & N. Ahituv,
Genomics 106(3),
September 2015,
Pages 159-164*

Massively parallel reporter assays (MPRAs)

Episomal reporter vector



Massively parallel reporter assay



Also known from other types of reporter gene assays, like Luciferase assays (e.g., promoters, microRNAs)

Gasparini et al, *Nat Reviews Genetics* 2020

MPRA "every way"

nature methods

ANALYSIS

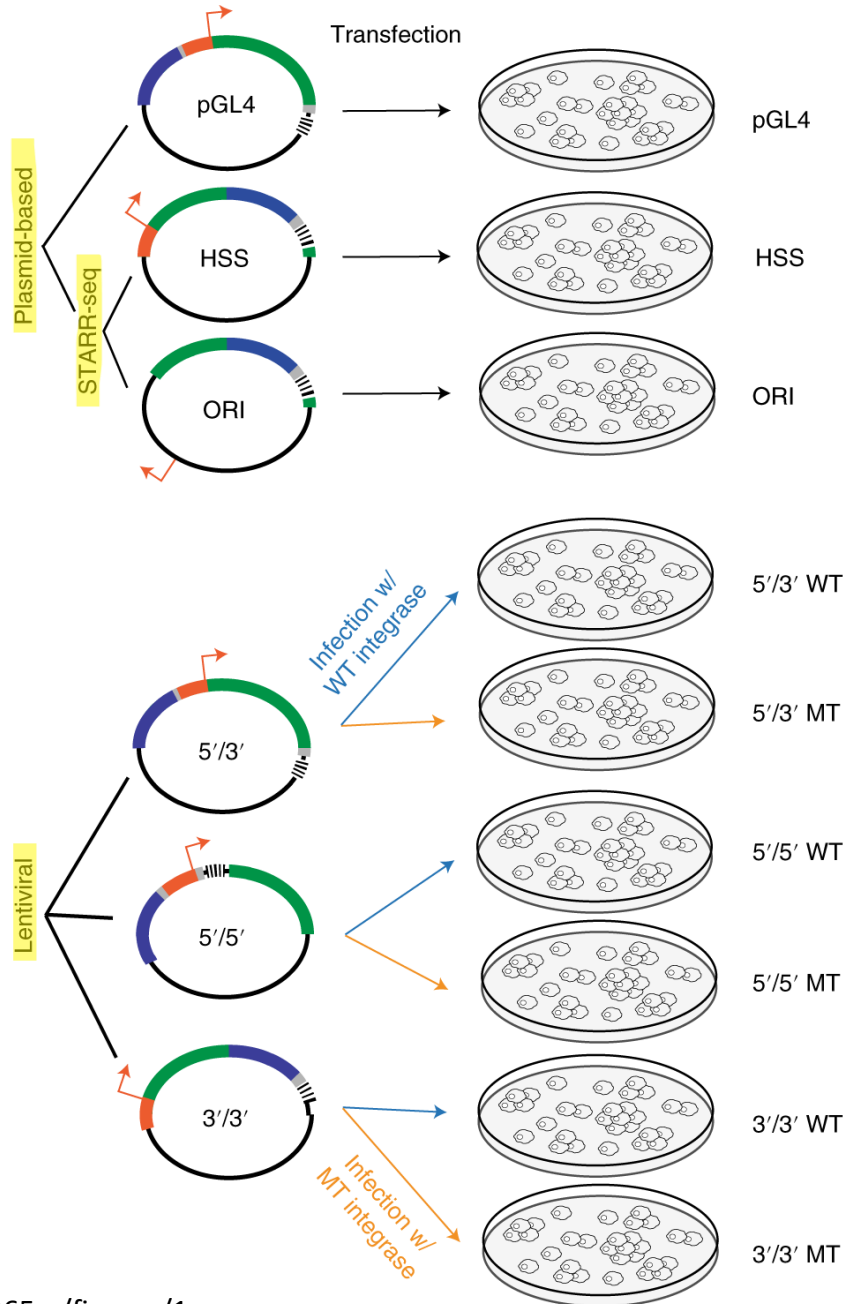
<https://doi.org/10.1038/s41592-020-0965-y>

Check for updates

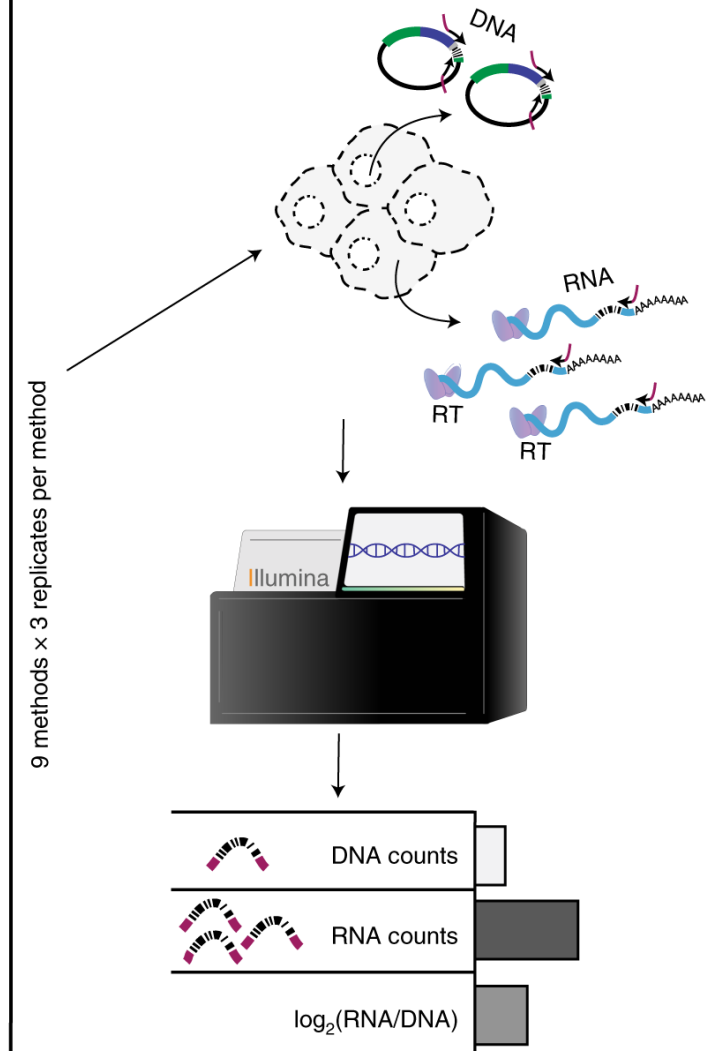
A systematic evaluation of the design and context dependencies of massively parallel reporter assays

Jason C. Klein^{1,9,11}, Vikram Agarwal^{1,2,11}, Fumitaka Inoue^{3,4,10,11}, Aidan Keith^{1,11}, Beth Martin¹, Martin Kircher^{1,5,6}, Nadav Ahituv^{3,4} and Jay Shendure^{1,7,8}

Massively parallel reporter assays (MPRAs) functionally screen thousands of sequences for regulatory activity in parallel. To date, there are limited studies that systematically compare differences in MPRA design. Here, we screen a library of 2,440 candidate liver enhancers and controls for regulatory activity in HepG2 cells using nine different MPRA designs. We identify subtle but significant differences that correlate with epigenetic and sequence-level features, as well as differences in dynamic range and reproducibility. We also validate that enhancer activity is largely independent of orientation, at least for our library and designs. Finally, we assemble and test the same enhancers as 192-mers, 354-mers and 678-mers and observe sizable differences. This work provides a framework for the experimental design of high-throughput reporter assays, suggesting that the extended sequence context of tested elements and to a lesser degree the precise assay, influence MPRA results.

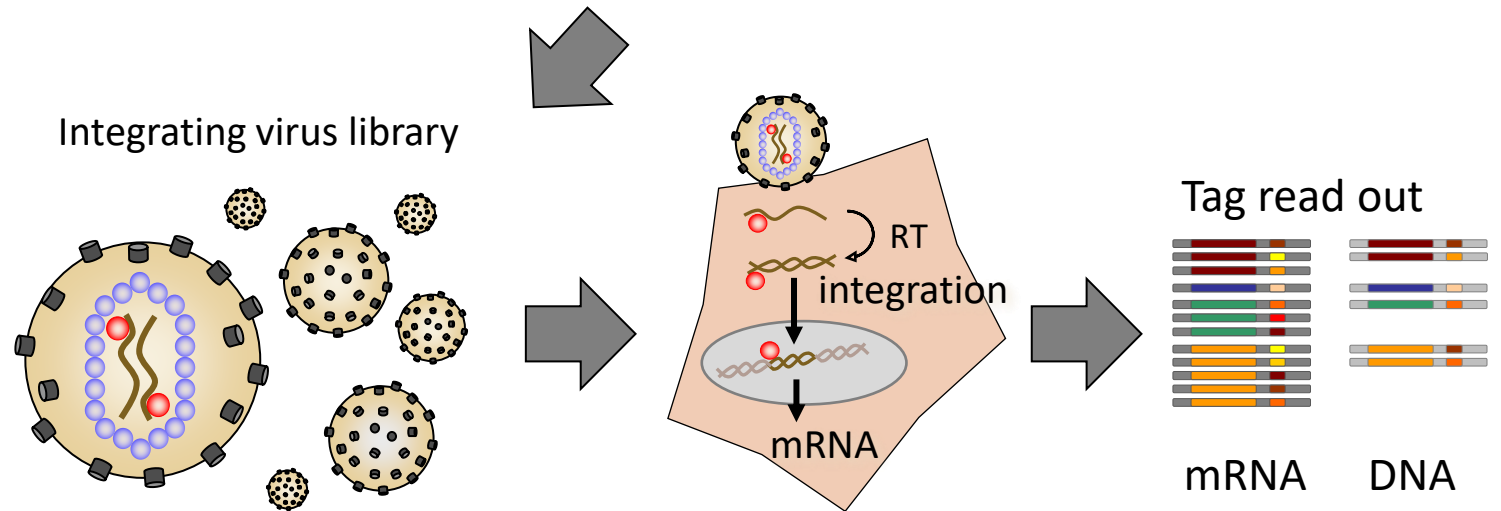
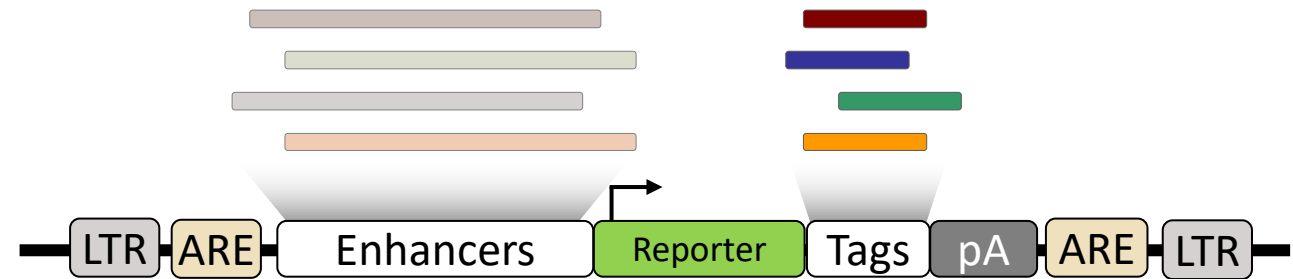
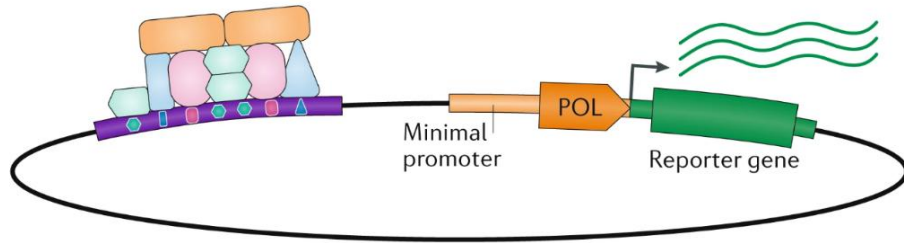


	Minimal promoter		2,440-element library
	Reporter		Barcode
	Spacer		TSS



Episomal vs. lentiMPRA genomic integration

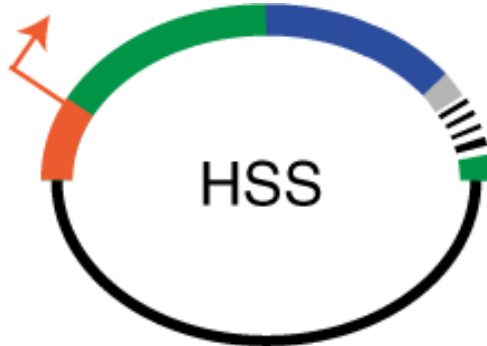
Episomal reporter vector



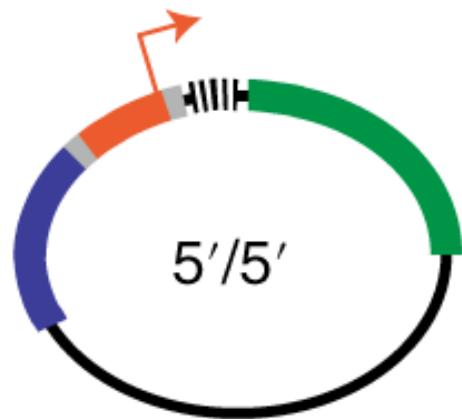
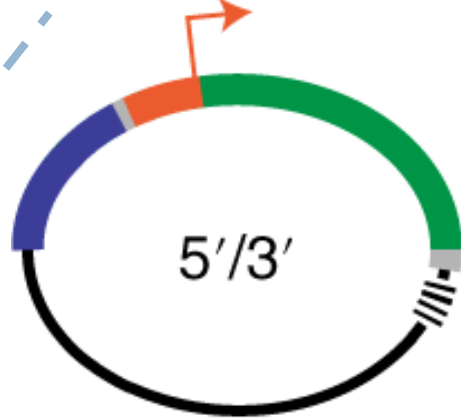
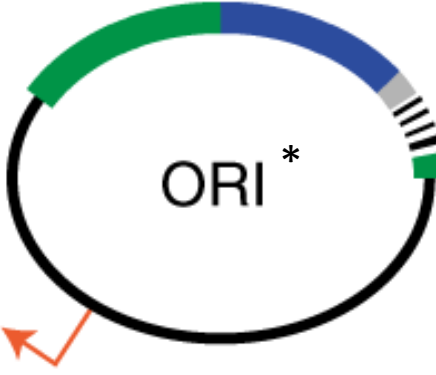
STARR-seq vs barcoded constructs

	Minimal promoter		2,440-element library
	Reporter		Barcode
	Spacer		TSS

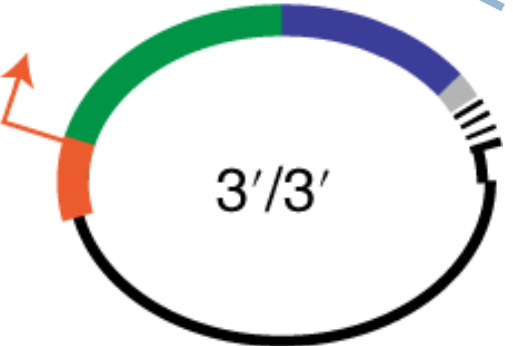
* <https://pubmed.ncbi.nlm.nih.gov/29256496/>

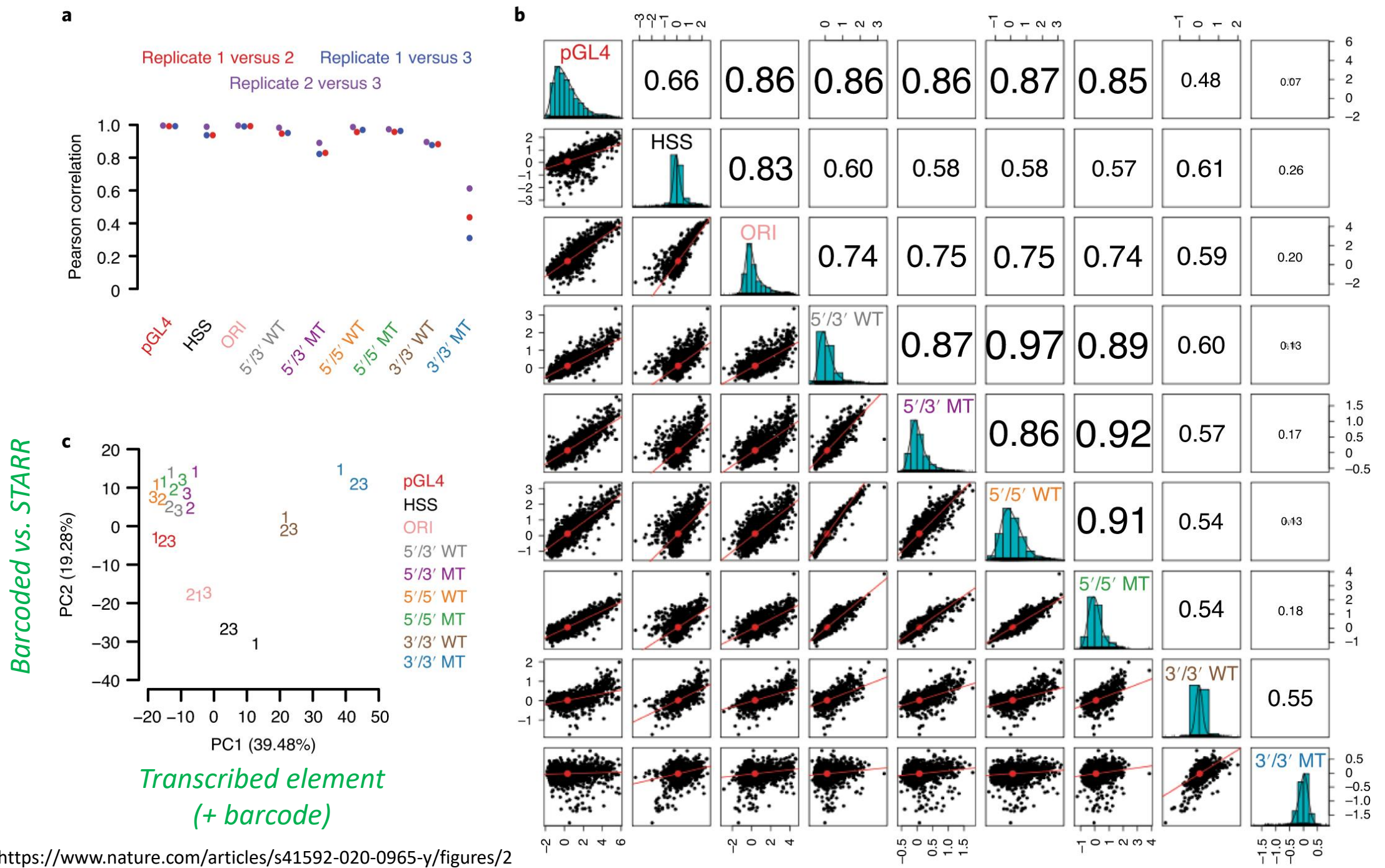


STARR-seq – tested sequence is being transcribed and identified through RNA sequencing

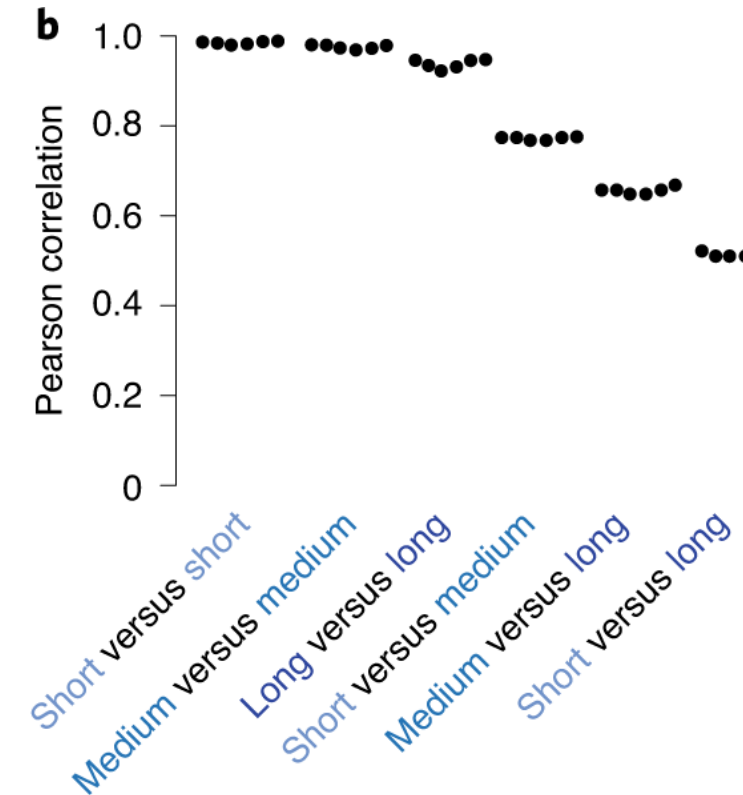
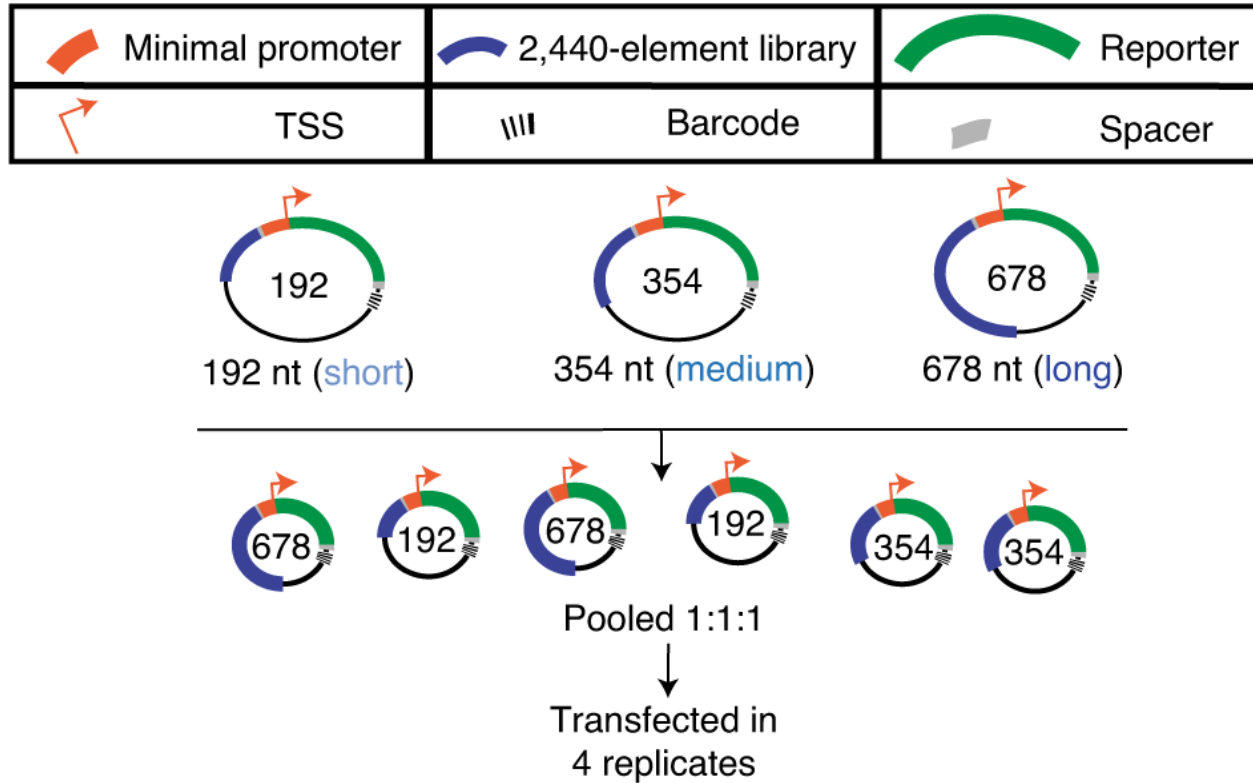


CRE-seq/barcoded constructs – tested sequence is represented by a (short) transcribed barcode





Length of the tested sequence has an effect



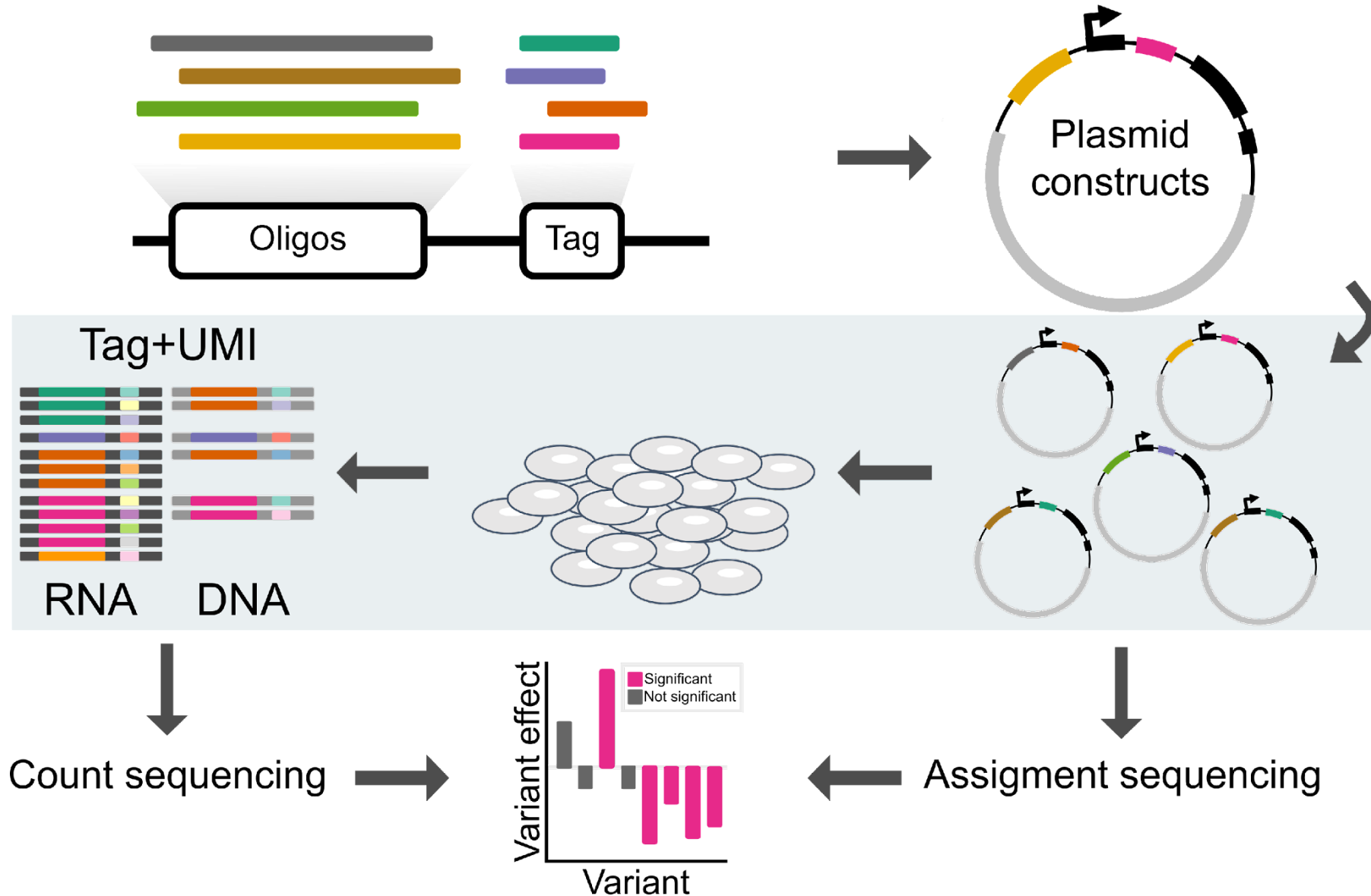
Testing longer sequences comes with more replicate variation, but longer sequences are adding biologically relevant signal.

Overview of challenges with MPRA

- Artificial nature (e.g., reporter, cell lines, ...)
- Experimental complexity (e.g., RNA/DNA extraction, working with complex DNA libraries, concordance between experimentalists/labs...)
- Variety of experimental designs (e.g., vector, targets, barcodes, UMIs, ...)
- Typically, length of tested sequences is constrained, e.g. by synthesis
- Elements are easier than variants
 - Variants require successful measurements for at least two alleles
 - Detection of variant effect will depend on activity of its element
 - Element effects can be modeled much better – reaching MPRA replicate correlations
- The more sequences (the higher the complexity) the more difficult the experiment

Part II – What's the output and how to process the data?

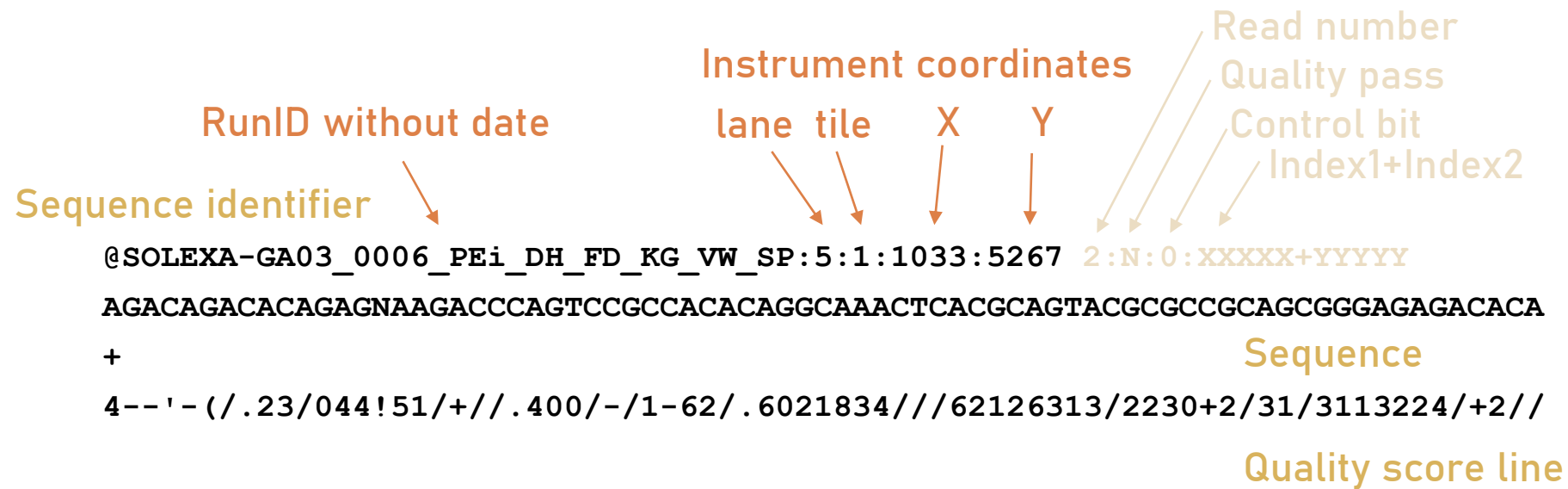
Massive Parallel Reporter Assay (MPRA)



FastQ files

```
@SOLEXA-GA03_0006_Pei_DH_FD_KG_VW_SP:5:1:1033:5267
AGACAGACACAGAGNAAGACCCAGTCCGCCACACAGGCAAACCTCACGCAGTACGCGCCGCAGCGGGAGAGACACAAACCGTAGTGCCTCTGCCTGTGTCTCTCTGCCTCTGTCTCTGTCTCTGTCTCTGTCTCTGCC
+
4--'-(/.23/044!51/+//.400/-/1-62/.6021834///62126313/2230+2/31/3113224/+2//.938643153256956368576789;979844.;797786984778:8:68595789943:5.:81:0885/61334/5+10--
@SOLEXA-GA03_0006_Pei_DH_FD_KG_VW_SP:5:1:1033:18417
CAGCATGCTGGTCGNCCCACCTCAGGGCCGAAGGCACCCATGCACCTGGCAGGTTGGCGTCCCCAGTGCTGAGGAAATGGTATCATTATGATCTGGATCTATGGAGGCGCCTTCCTCATGGGGTCCGGCCACGGGGCCAACTTCCTCAACAACCTACCT
+
45628576753499!37447349;726458:82849745;;998669:46:96566336533336723.2133221536543752688898;=9.;8<=9<==96.;87988:97697;;6676645836486755514960863340461462540.0
```

⋮



Phred* quality score

Given e , the estimated probability of the base call being wrong:

$$Q = -10 \log_{10} e$$

Higher Q scores indicate a smaller probability of error

Example: A quality score of 20 represents an error rate of 1 in 100, i.e., a call accuracy of 99%; a quality score of 30 corresponds to an error rate of 1/1000 (accuracy of 99.9%)

Encoding usually as ASCII char 33 to 74 (depending on technology) for a Phred quality score from 0 to 41.

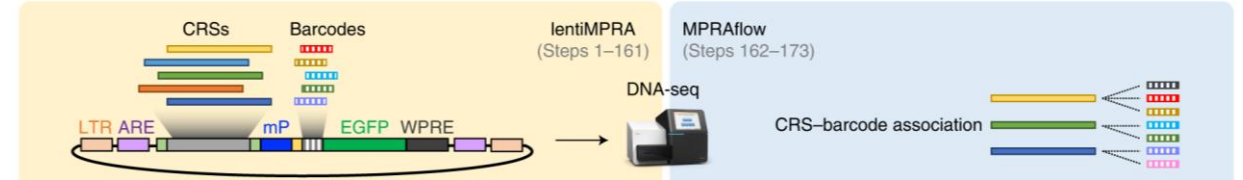
Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
32	20	Space	64	40	@	96	60	`
33	21	!	65	41	A	97	61	a
34	22	"	66	42	B	98	62	b
35	23	#	67	43	C	99	63	c
36	24	\$	68	44	D	100	64	d
37	25	%	69	45	E	101	65	e
38	26	&	70	46	F	102	66	f
39	27	'	71	47	G	103	67	g
40	28	(	72	48	H	104	68	h
41	29	)	73	49	I	105	69	i
42	2A	*	74	4A	J	106	6A	j
43	2B	+	75	4B	K	107	6B	k
44	2C	,	76	4C	L	108	6C	l
45	2D	-	77	4D	M	109	6D	m
46	2E	.	78	4E	N	110	6E	n
47	2F	/	79	4F	O	111	6F	o
48	30	0	80	50	P	112	70	p
49	31	1	81	51	Q	113	71	q
50	32	2	82	52	R	114	72	r
51	33	3	83	53	S	115	73	s
52	34	4	84	54	T	116	74	t
53	35	5	85	55	U	117	75	u
54	36	6	86	56	V	118	76	v
55	37	7	87	57	W	119	77	w
56	38	8	88	58	X	120	78	x
57	39	9	89	59	Y	121	79	y
58	3A	:	90	5A	Z	122	7A	z
59	3B	;	91	5B	[	123	7B	{
60	3C	<	92	5C	\	124	7C	
61	3D	=	93	5D	]	125	7D	}
62	3E	>	94	5E	^	126	7E	~
63	3F	?	95	5F	_	127	7F	□

* Named after a software that introduced quality score calculation for Sanger sequencing

Output data from association or assignment sequencing

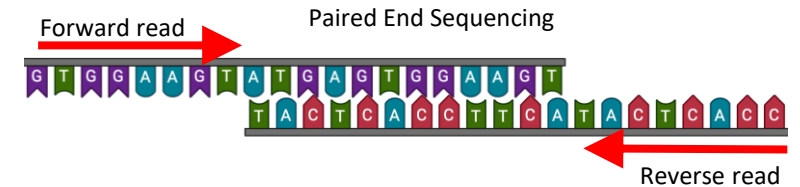
Main sequencing reads

- Single or Pair of FASTQ file(s) of the target



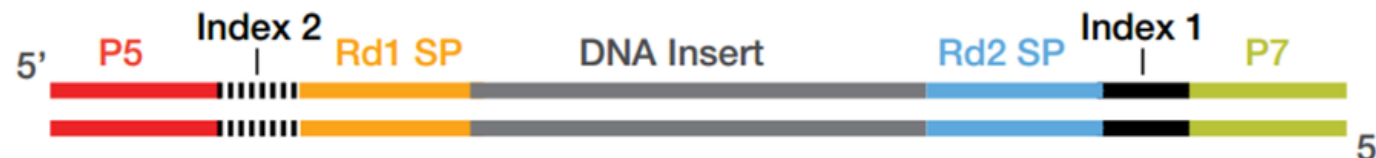
Target barcode

- Typically, separate FASTQ file (for i5 and/or i7 index)



Sample index

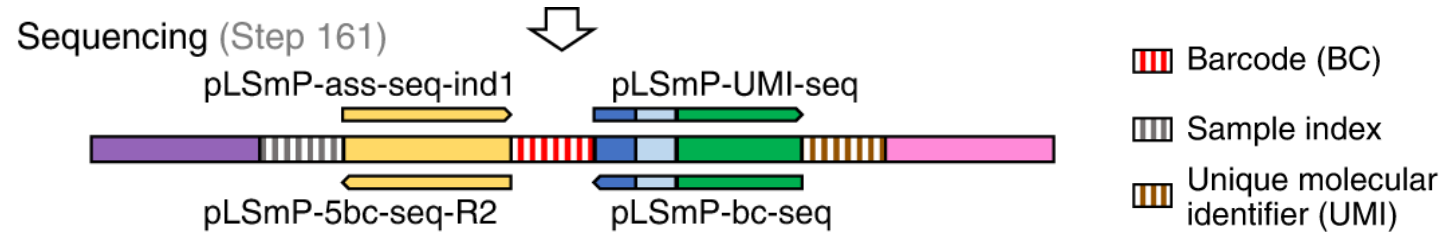
- Typically, separate FASTQ file (for i5 and/or i7 index), most likely already demultiplexed by sequencing core



<https://support.illumina.com/bulletins/2020/06/illumina-adapter-portfolio.html>

Index, barcode and UMI sequences

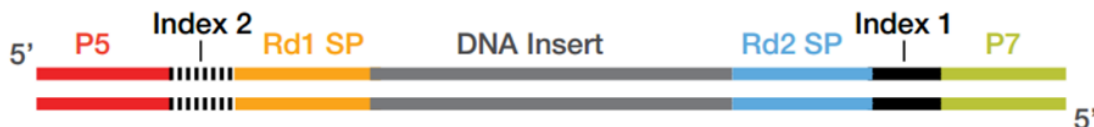
TAACAGCCAA
CTAAGAGTCC
TTACTGCCTT
CGCTGAATTC
TGACGTCCTT
TGTGTGTAAC
AGATTGAGAG
GATAATGATG
TTGAAGAAGC
CTGATACTTC



... are short sequences used to uniquely tag a sample, partition/cell/targets or single molecules in an experiment

Typically,

- Indexes tag samples (e.g., replicates)
- Barcodes tag targets (e.g., counts)
- UMIs (unique molecular identifiers) tag molecules before amplification (for count estimation and error correction)



<https://support.illumina.com/bulletins/2020/06/illumina-adapter-portfolio.html>

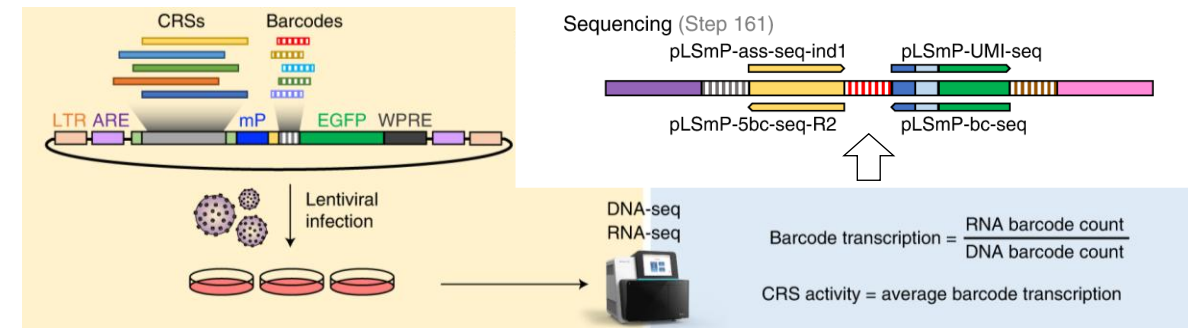
Output data from count sequencing

Main sequencing reads

- Single or Pair of FASTQ file(s) of barcode sequences

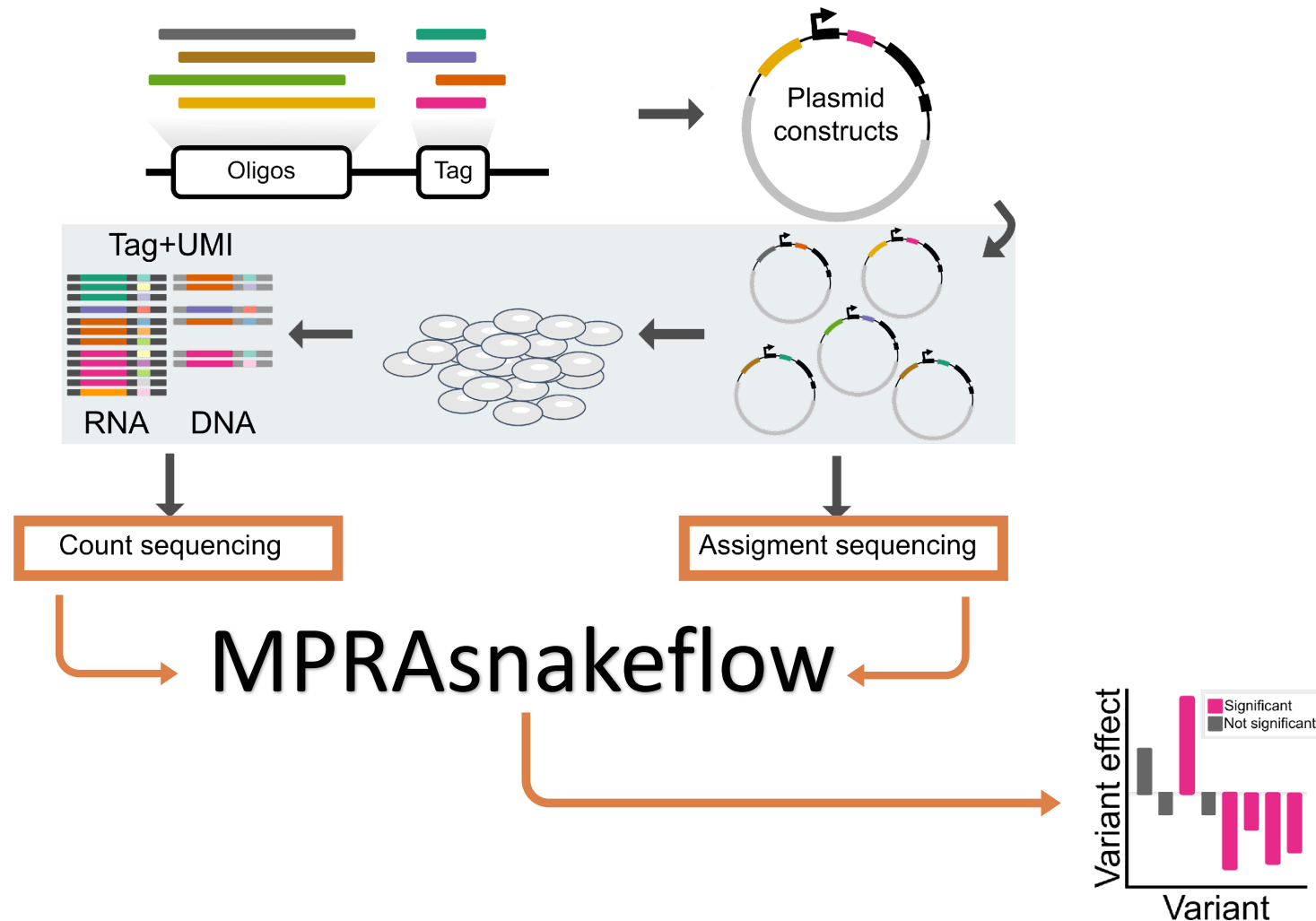
Sample index and or UMI reads

- Typically, separate FASTQ files
- OR
- Part of first/second read in main output file (Barcode trimming step necessary)



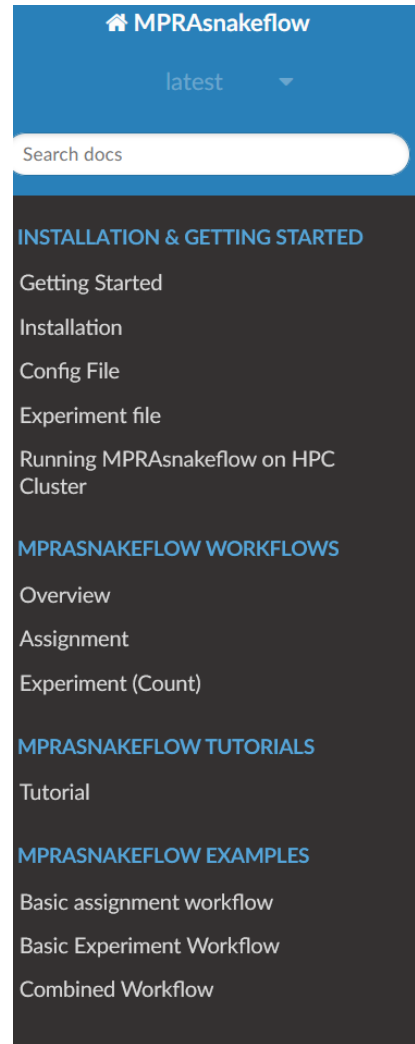
<https://support.illumina.com/bulletins/2020/06/illumina-adapter-portfolio.html>

MPRAsnakeflow – Pipeline to process sequencing data from MPRA to create count tables for candidate sequences and variants



MPRAsnakeflow

<https://github.com/kircherlab/MPRAsnakeflow>



[MPRAsnakeflow's Documentation](#)

[View page source](#)

MPRAsnakeflow's Documentation

snakemake `≥8.24.1` conda `>24.7.1`

Welcome!

MPRAsnakeflow is a pipeline designed to process sequencing data from Massively Parallel Reporter Assays (MPRAs) to create count tables for candidate sequences tested in the experiment.

MPRAsnakeflow is built on top of [Snakemake](#) (version `≥8.24.1` required) and is configured via a `.yaml` file.

Authors

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Usage

If you use this workflow in a paper, don't forget to give credit to the authors by citing the URL of the (original) repository.

Installation & Getting Started

<https://mprasnakeflow.readthedocs.io>

Technical details

- **MPRAsnakeflow is implemented in snakemake**
 - Python based workflow management software
 - <https://snakemake.readthedocs.io>
- **Software packages managed via conda or apptainer**
- **Input controlled via config files (YAML)**

Possible Snakemake commands

```
# basic command
snakemake --sdm conda --cores 4 --configfile config.yaml

# run within working dir
snakemake --sdm conda -c 4 --configfile config.yaml \
--snakefile path/to/MPRAsnakeflow/workflow/Snakefile

# use different working dir (run in MPRAsnakeflow)
snakemake --sdm conda -c 4 --configfile config.yaml \
--directory path/to/working/dir
```

Config file example

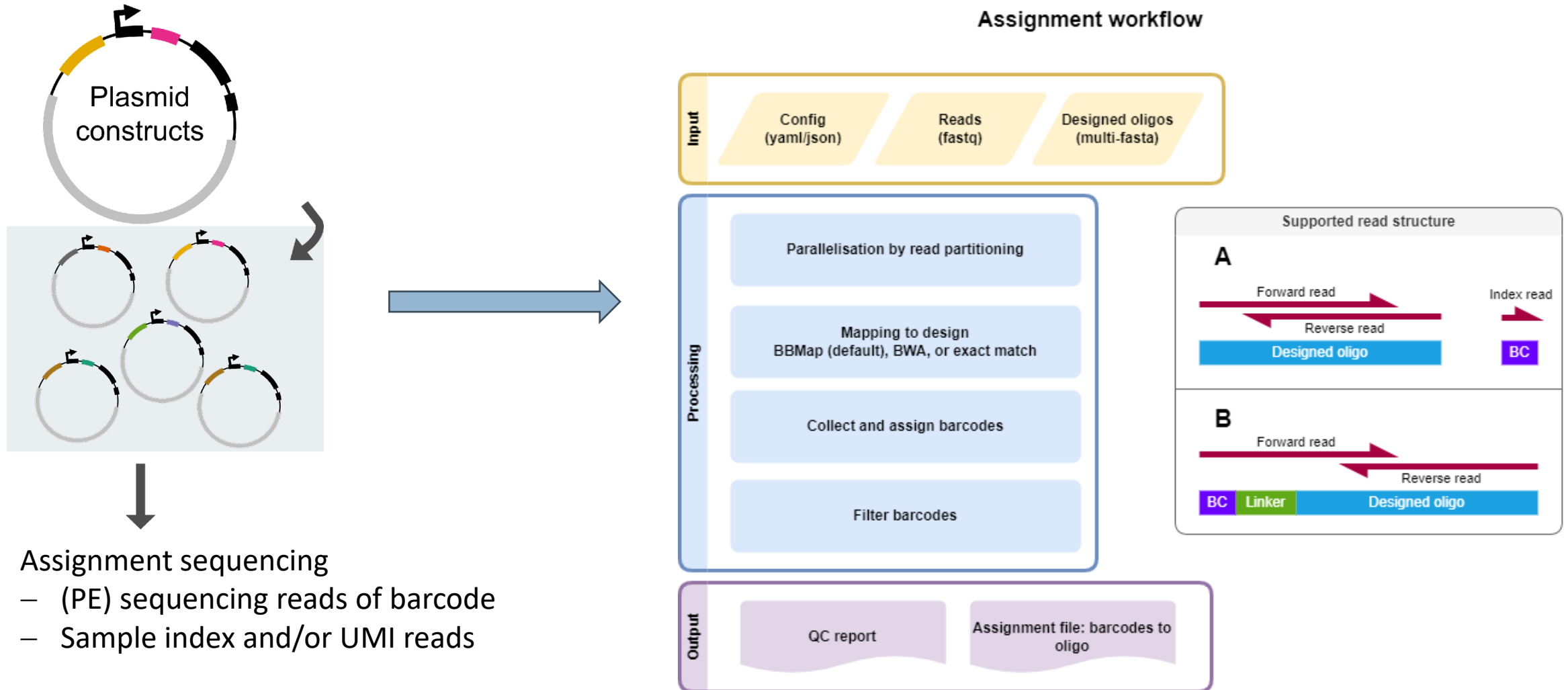
```
assignments:
  exampleAssignment: # name of your analysis
    bc_length: 15
    alignment_tool:
      tool: bbmap
      configs:
        sequence_length: 270
        alignment_start: 1
    FW: [resources/Assignment_Basic/R1.fastq.gz]
    BC: [resources/Assignment_Basic/R2.fastq.gz]
    REV: [resources/Assignment_Basic/R3.fastq.gz]
    design_file: resources/design.fa
    configs:
      default: {} # name of additional filtering
```

MPRAsnakeflow

How?

- Assignment → assign barcodes to corresponding tested elements (optional)
- Experiment → process sequenced barcodes to create count tables

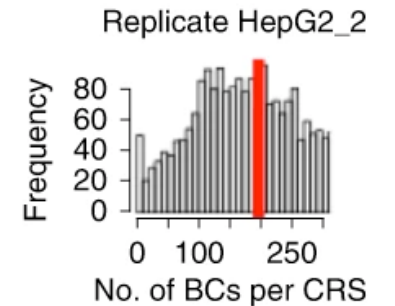
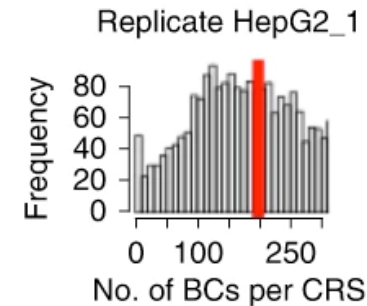
Assignment Workflow: Overview



MPRAsnakeflow: assignment workflow

Why?

- Require correct assignment of barcodes to sequences
 - Mis-assignment could lead to misinterpretation of element activity, or worse, skewing of a variant log ratio
- Look for:
 - Sufficient coverage of BC / seq
 - Avoid BC mapping to multiple sequences
 - Measure library complexity and achieve close to complete coverage in association sequencing



Outlook hands-on: Association of Barcode/Tag Sequences



03 Standard Pipeline for MPRA Analysis: MPRAsnakeflow

External links:

- [MPRAsnakeflow repository](#)
- [MPRAsnakeflow documentation](#)
- [MPRAsnakeflow tutorial](#)

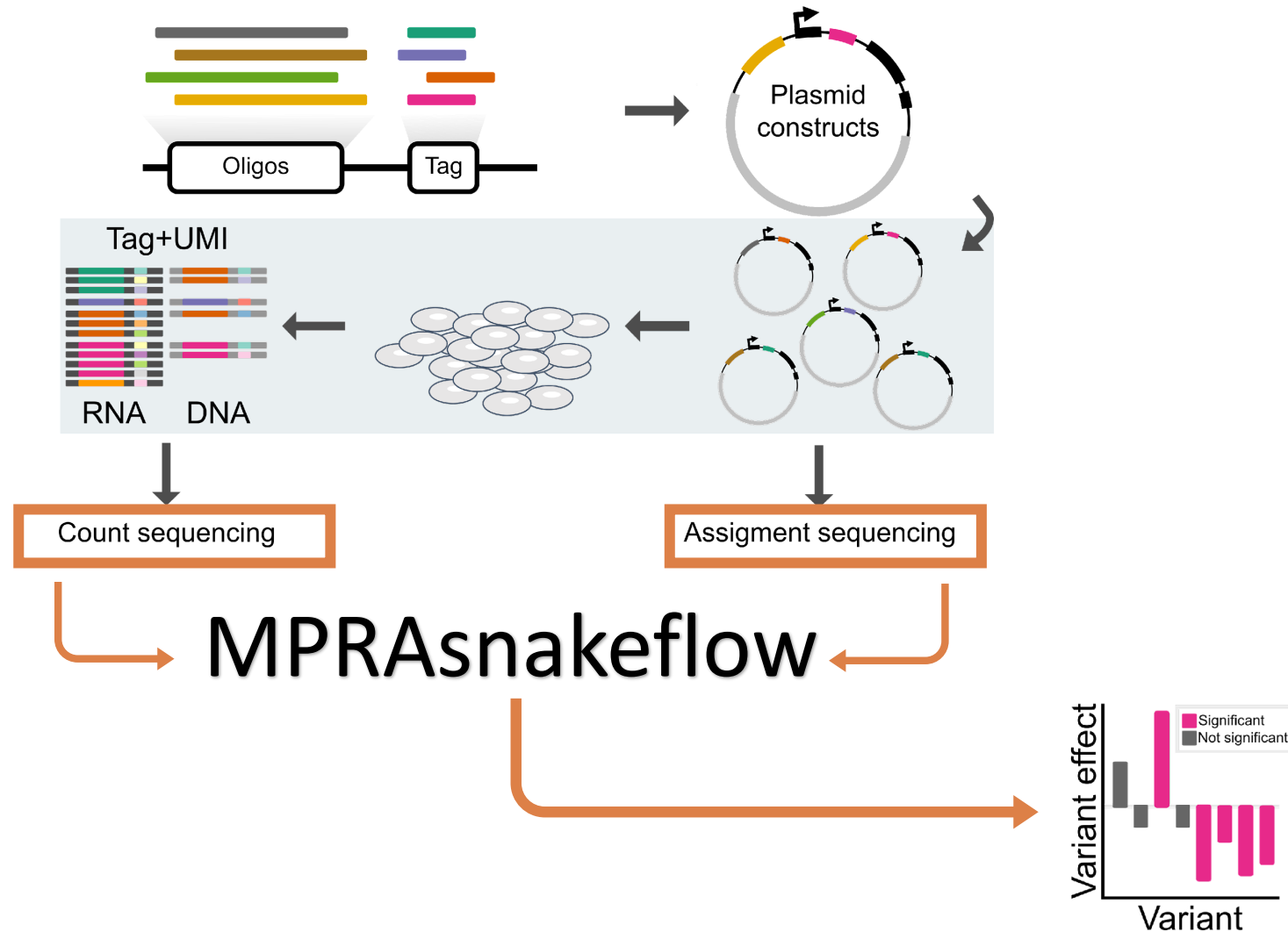
Hands-on:

- Association of Barcode/Tag Sequences: [Jupyter/Colab notebook](#)
- Count Sequencing Analysis: [Jupyter/Colab notebook](#)

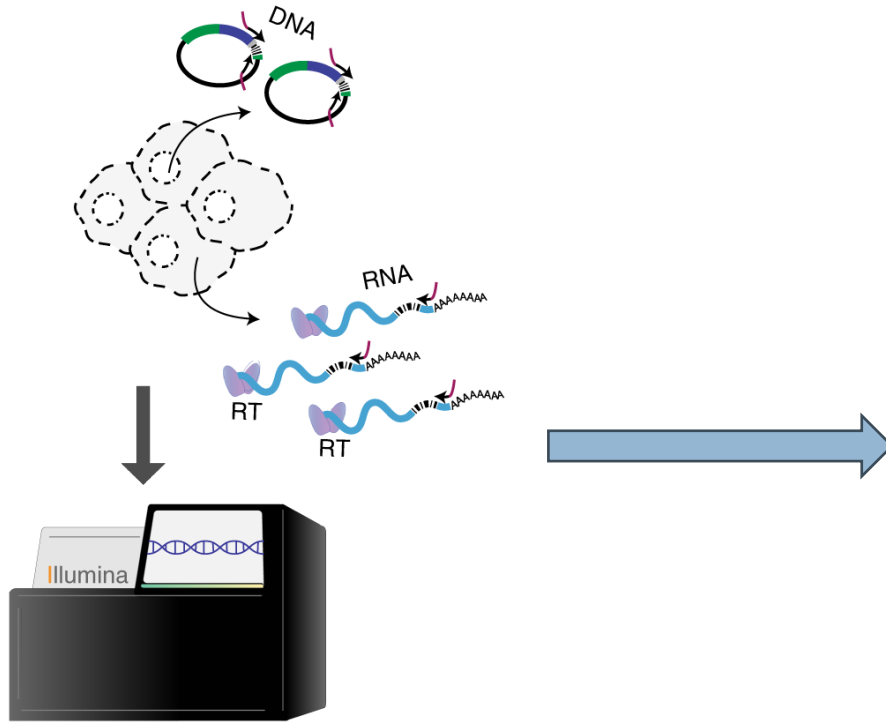


https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

Massive Parallel Reporter Assay (MPRA)



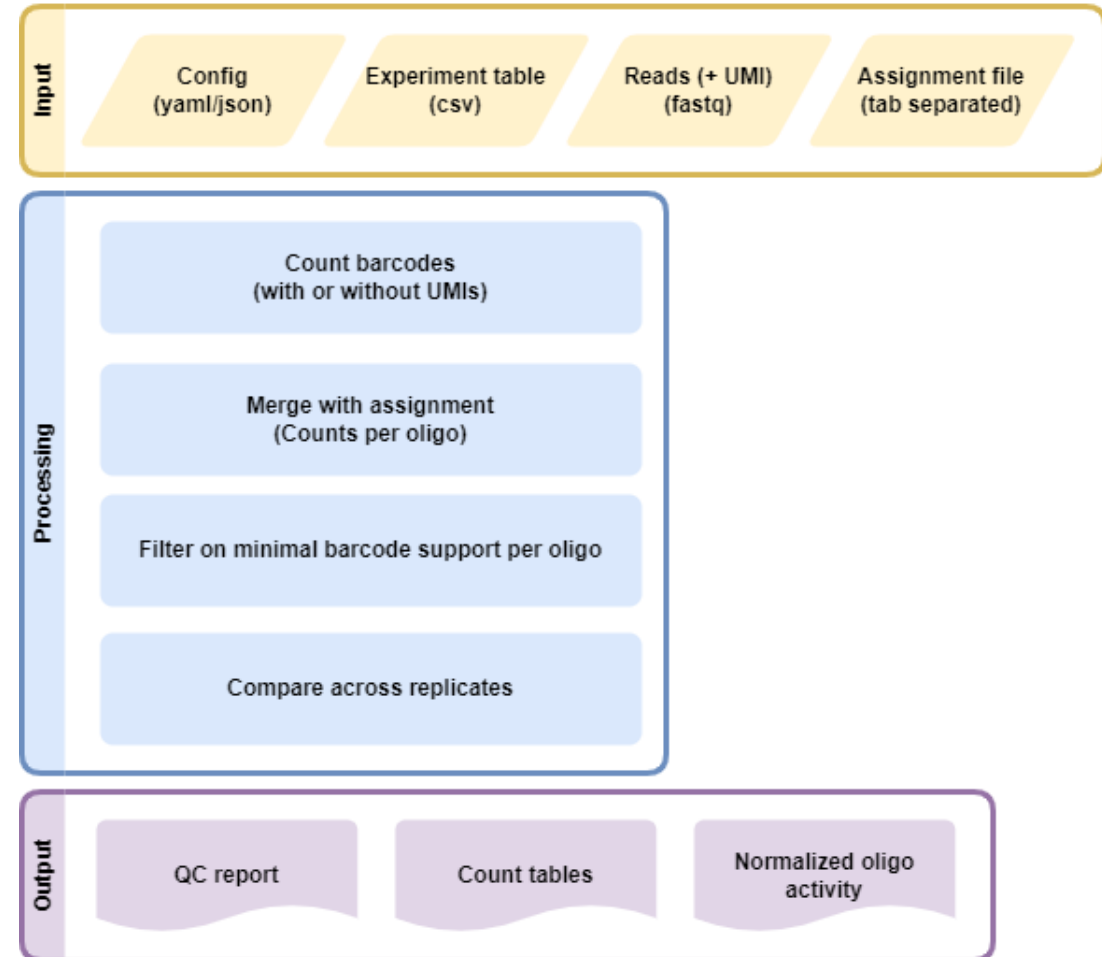
Experiment Workflow: Overview



Assignment sequencing

- Sequencing reads of plasmid insert
- Target barcode
- Sample index

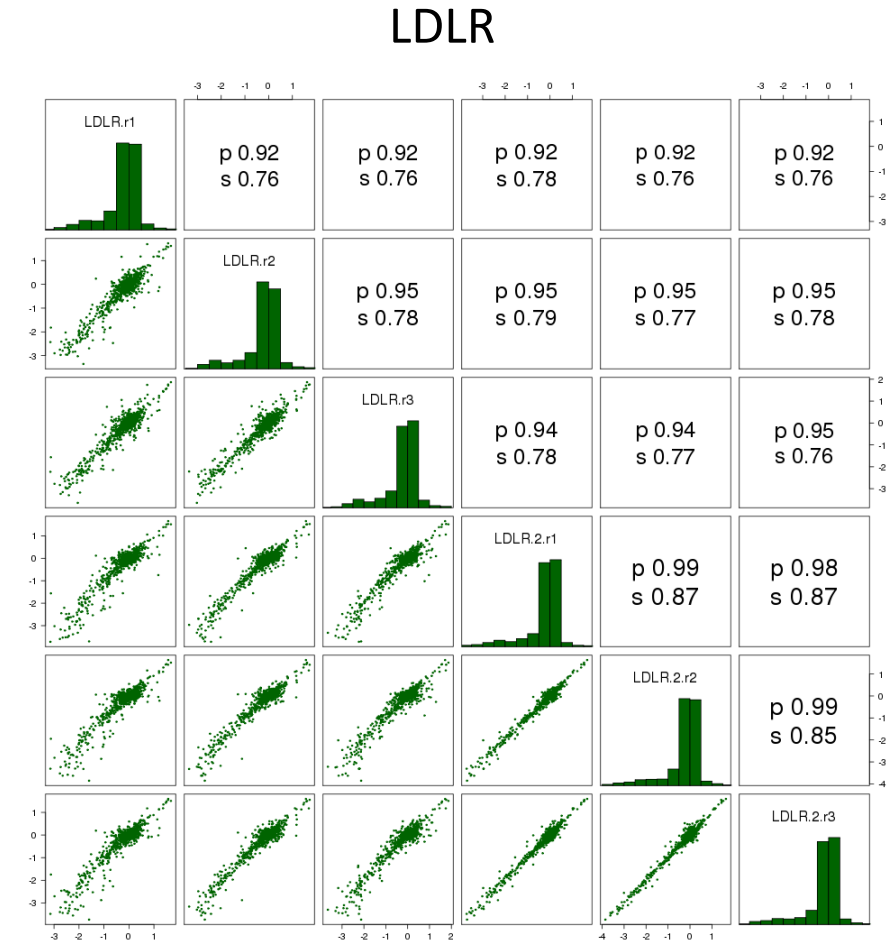
Experiment workflow



MPRAsnakeflow: Count workflow

Why?

- Need to summarize and integrate data
 - Each tested element is represented by many barcodes – information stored in the assignment
 - Barcode sequence might influence measured activity
 - Variants are represented by at least two elements
- Look for:
 - Sufficient coverage of elements with barcodes
 - Good correlation between replicates



Combined replicates: p 0.97 / s 0.84

<https://www.nature.com/articles/s41467-019-11526-w>

(Saturation mutagenesis data)

Outlook hands-on: Count Sequencing Analysis



03 Standard Pipeline for MPRA Analysis: MPRAsnakeflow

External links:

- [MPRAsnakeflow repository](#)
- [MPRAsnakeflow documentation](#)
- [MPRAsnakeflow tutorial](#)

Hands-on:

- Association of Barcode/Tag Sequences: [Jupyter/Colab notebook](#)
- Count Sequencing Analysis: [Jupyter/Colab notebook](#)

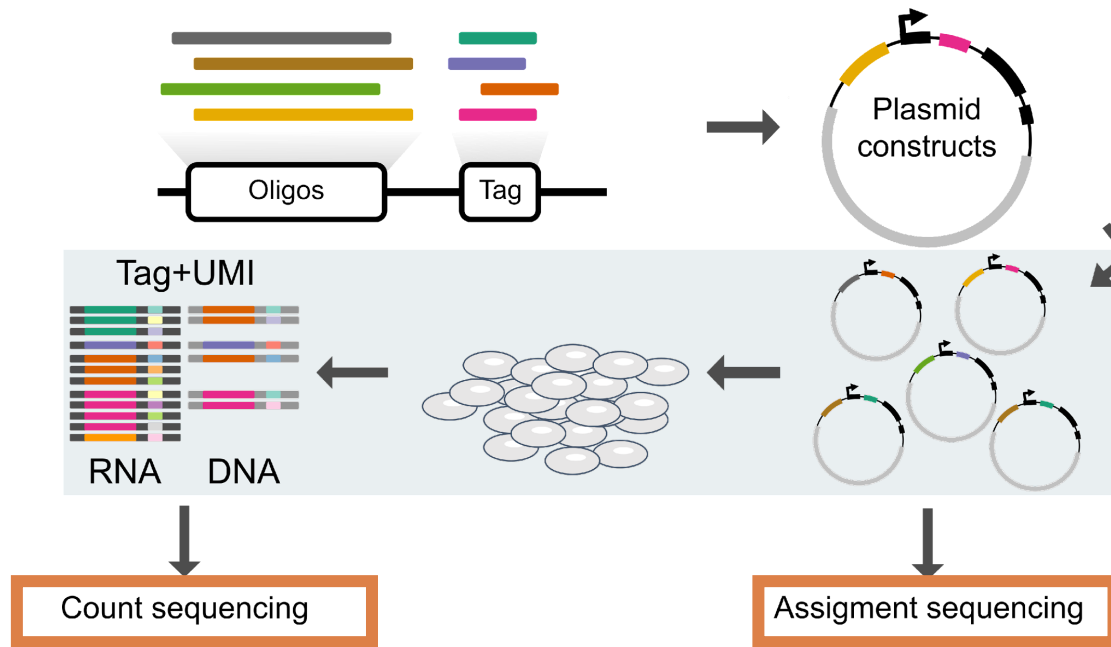


https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

How to continue from a count matrix?

ID	Barcode	DNA1	DNA2	DNA3	DNA4	DNA5	RNA1	RNA2	RNA3	RNA4	RNA5
Oligo1	ACTCGAAACCAGGTCCCATT	9	9	5	16	6	6	7	13	8	18
Oligo1	ACTGGACAATTTAATAGCAT	1	0	0	0	1	0	0	0	0	3
Oligo1	ACTTTATGCAACACGTAAAG	1	1	0	0	1	1	1	0	0	1
Oligo1	AGAACATATCTGGCACATGA	23	14	18	5	13	11	36	3	35	32
Oligo1	AGACGGCTTACGAGGCATCC	16	8	15	13	11	20	29	14	15	19
Oligo1	AGCAACAACAGGTCAACCAG	24	24	22	15	18	2610	3830	4760	1215	1935
Oligo1	AGGGGGGAGTACCGAGTGAAT	2	7	0	3	2	1	7	2	5	7
Oligo1	AGTAAACACCGCATTATGGC	8	5	3	7	1	2	7	5	8	3
Oligo1	AGTAGCGTAAACCTCCCCCG	9	6	9	7	11	9	23	3	16	15
Oligo1	AGTCCATCTGAATACACGTA	21	13	22	14	26	14	50	31	33	31
Oligo1	ATACCTTACTAACACAACCTT	7	3	3	5	7	2	15	3	4	5
Oligo1	ATATATCAAACGCGGATACA	2	6	4	2	7	5	12	6	12	12
Oligo1	ATCAGTACATTGTCACATAT	6	4	2	1	5	6	8	4	15	12
Oligo1	ATCGTCTTCCCGCAAAGGCA	4	2	7	5	1	1	3	3	6	8
Oligo1	ATCGTTAGGGCAAGGAGCAC	37	21	40	23	38	31	116	67	53	64

Statistical analysis of counts and identifying significant variant and element effects

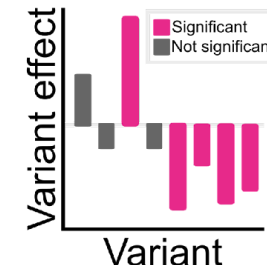


$$\text{DNA} \sim \text{Gamma}(k, b)$$

$$\text{RNA}|\text{DNA} \sim \text{Poisson}(\alpha \cdot \text{DNA})$$

$$\text{RNA} \sim \text{NB} \left(\mu = \frac{\alpha \cdot k}{\beta}, \psi = k \right)$$

BCalm, mpralm, MPRAnalyze, ...



Outlook hands-on: Data Analysis Steps (Regions and Variants)



04 MPRA Data Analysis

Hands-on:

- Data Analysis Steps (Regions and Variants): [README](#), [Jupyter/Colab notebook](#)

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

Thank You! Questions?

kircherlab.github.io

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UNIVERSITÄTSKLINIKUM
Schleswig-Holstein

IBIH Berlin Institute
of Health
@Charité

The background of the image is a dense field of coffee beans, likely Arabica, which are dark brown and have a glossy, slightly wrinkled texture. The beans are scattered across the entire frame, creating a rich, textured backdrop. The lighting is soft and even, highlighting the natural curves and creases of the beans.

COFFEE

BREAK

